DETAILS

NAME :THOTA YEDUKONDALU

SECTION :CSE 1

ROLL NO :65

Case 1: University Scholarship Allocation

Problem Statement

Evaluate applicants using aggregate marks, subject wise minimum marks, attendance, and family income. Rank eligible students and apply predefined tie breaking rules.

Objective

Identify eligible applicants and rank them by total marks; break ties by attendance, then lower family income.

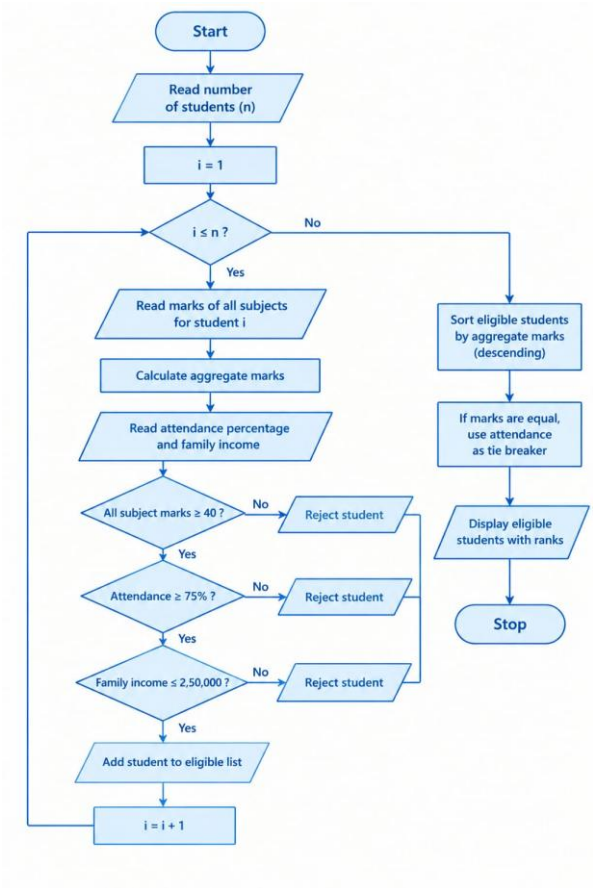
Assumptions

- Subjects: Math, English, Science.
- The user enters each subject's minimum mark, minimum attendance, and maximum income.
- Applicants must meet **all** criteria.
- Ties are resolved by higher attendance, then lower income, then name alphabetically.

Algorithm

1. Read the eligibility thresholds and number of applicants.
2. For each applicant, read marks, attendance, and family income.
3. Check that every subject mark and both other criteria meet the thresholds.
4. Store eligible applicants.
5. Sort eligible applicants by the tie breaking rules.
6. Display the ranked list.

Flowchart



Pseudocode

case1.py - C:/Users/Deepthi/OneDrive/Desktop/activity/case1.py (3.13.14)

File Edit Format Run Options Window Help

```
print("UNIVERSITY SCHOLARSHIP ALLOCATION")

students = []

n = int(input("Enter number of students: "))

for i in range(n):
    print(f"\nStudent {i + 1}")

    name = input("Enter student name: ")

    mark1 = float(input("Enter marks in Subject 1: "))
    mark2 = float(input("Enter marks in Subject 2: "))
    mark3 = float(input("Enter marks in Subject 3: "))

    attendance = float(input("Enter attendance percentage: "))
    income = float(input("Enter annual family income: ₹"))

    aggregate = (mark1 + mark2 + mark3) / 3

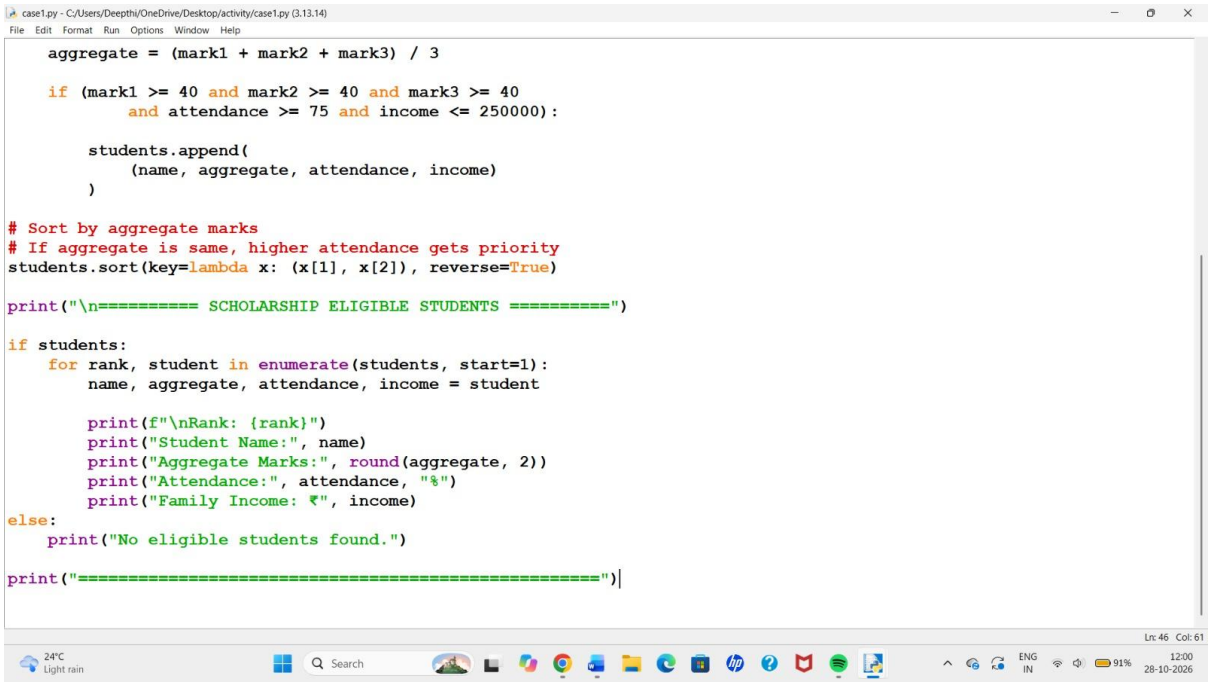
    if (mark1 >= 40 and mark2 >= 40 and mark3 >= 40
            and attendance >= 75 and income <= 250000):

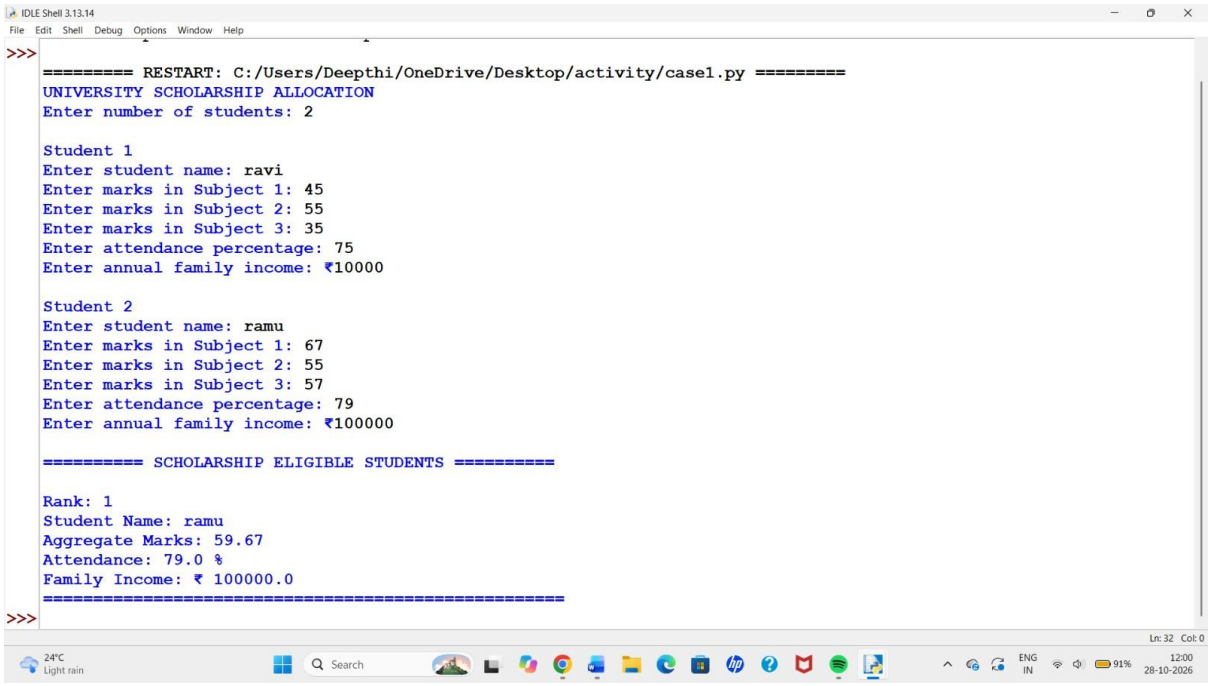
        students.append(
            (name, aggregate, attendance, income)
        )

# Sort by aggregate marks
# If aggregate is same, higher attendance gets priority
students.sort(key=lambda x: (x[1], x[2]), reverse=True)
```

Ln: 46 Col: 61

24°C Light rain
Search
ENG IN
91%
12:00 28-10-2026





INPUT subject minimums, attendance minimum, income maximum

INPUT number of applicants

eligible = empty list

FOR each applicant

INPUT name, subject marks, attendance, income

IF all marks meet minimums AND attendance meets minimum

```
    AND income is within maximum
     ADD applicant to eligible
  ENDIF
ENDFOR

SORT eligible by marks descending, attendance descending,
   income ascending, name ascending
DISPLAY eligible
```

Python Code

```
subjects = ["Math", "English", "Science"]
minimums = [float(input(f"Minimum mark for {s}: ")) for s in subjects]
min_attendance = float(input("Minimum attendance (%): "))
max_income = float(input("Maximum family income: "))
students = []

count = int(input("Number of applicants: "))

for _ in range(count):
  print("\nApplicant details")
  name = input("Name: ")
  marks = [float(input(f"{s} mark: ")) for s in subjects]
  attendance = float(input("Attendance (%): "))
  income = float(input("Family income: "))

  meets_marks = all(marks[i] >= minimums[i] for i in range(len(subjects)))
  if meets_marks and attendance >= min_attendance and income <= max_income:
     students.append([name, sum(marks), attendance, income])
```

```
students.sort(key=lambda x: (-x[1], -x[2], x[3], x[0].lower()))


print("\nEligible applicants, ranked")
if not students:
  print("No eligible applicants.")
else:
  for rank, student in enumerate(students, start=1):
    print(
      f"{rank}. {student[0]} | Total marks: {student[1]:.1f} "
      f"| Attendance: {student[2]:.1f}% | Income: {student[3]:.2f}"
    )
```

Test Case and Expected Output

Assume minimum marks are 60 in each subject, attendance minimum is 75%, and income maximum is 300,000. Applicant Asha has marks 80, 75, 90, attendance 90%, income 250,000. Applicant Bala has marks 85, 85, 75, attendance 80%, income 280,000.

Eligible applicants, ranked

1. Asha | Total marks: 245.0 | Attendance: 90.0% | Income: 250000.00

2. Bala | Total marks: 245.0 | Attendance: 80.0% | Income: 280000.00

Conclusion

The program filters applicants against all the entered criteria and ranks eligible students using the specified tie breaking rules.

Case 2: Progressive Electricity Billing

Problem Statement

Calculate household electricity bills using consumption slabs, fixed charges, surcharges, and rebates. Compare bills across months and identify unusual increases.

Objective

Calculate each monthly bill and flag a bill that is more than 50% higher than the previous month.

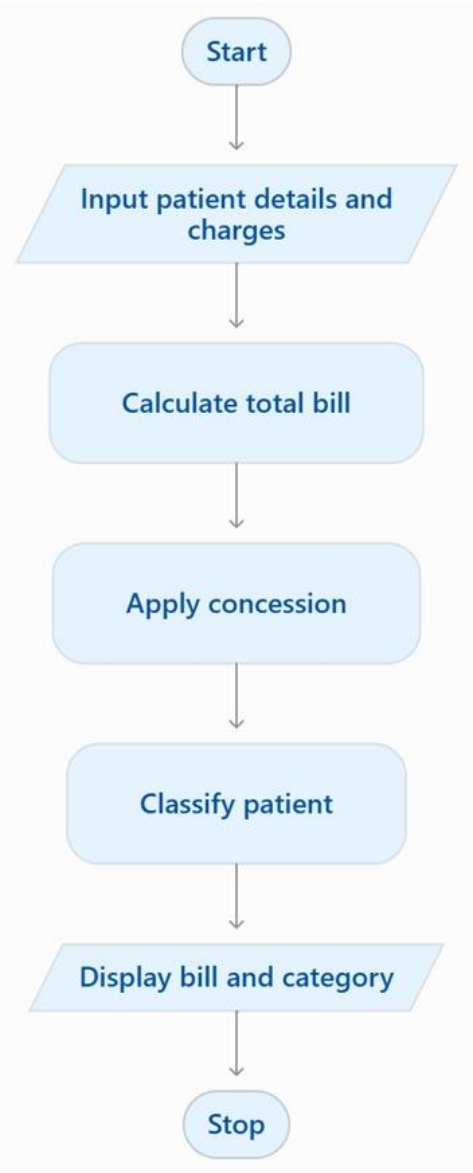
Assumptions

- First 100 units cost 2 per unit.
- Next 100 units cost 3 per unit.
- Units above 200 cost 5 per unit.
- Fixed charge is 50 per month.
- A 5% surcharge applies when the bill before surcharge exceeds 500.
- A 2% rebate applies when usage is below 100 units.
- An unusual increase means the current bill is more than 1.5 times the previous bill.

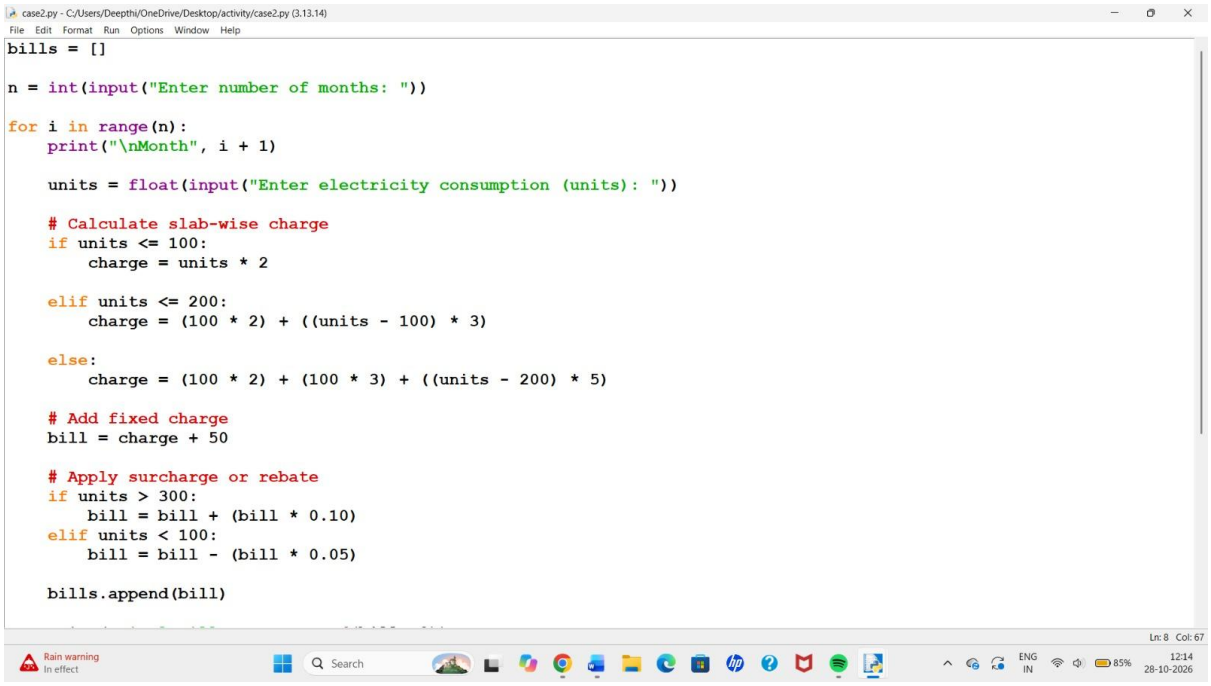
Algorithm

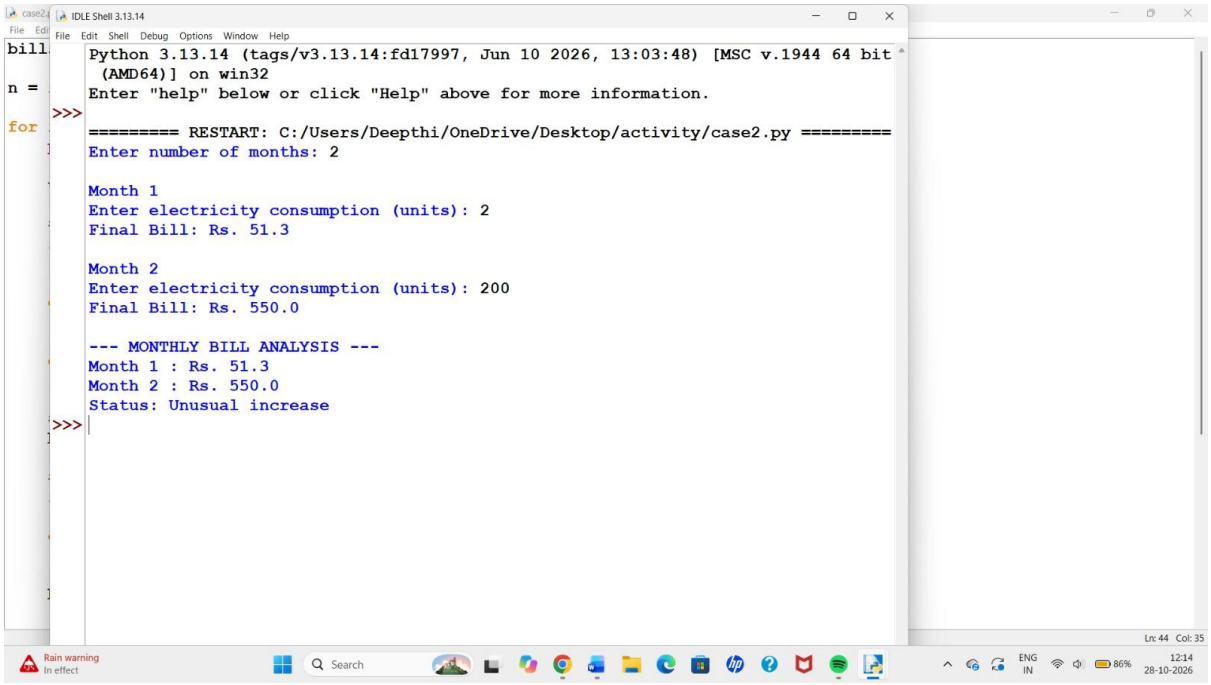
1. Read the number of months.
2. For each month, read units consumed.
3. Calculate the energy charge by slab.
4. Add the fixed charge and apply any surcharge or rebate.
5. Compare with the previous month and display a warning if the increase is unusual.

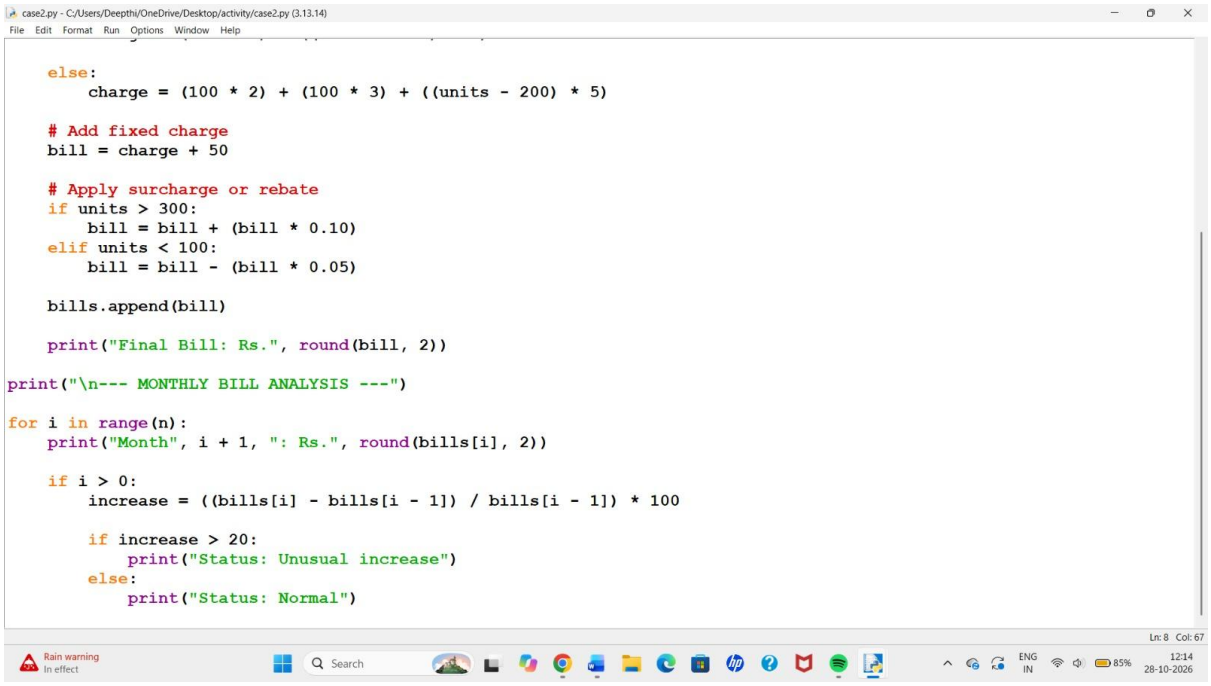
Flowchart


Start
Input patient details and charges
Calculate total bill
Apply concession
Classify patient
Display bill and category
Stop

Pseudocode


case2.py - C:/Users/Deepthi/OneDrive/Desktop/activity/case2.py (3.13.14)
File Edit Format Run Options Window Help
bills = []
n = int(input("Enter number of months: "))
for i in range(n):
print("\nMonth", i + 1)
units = float(input("Enter electricity consumption (units): "))
Calculate slab-wise charge
if units <= 100:
charge = units * 2
elif units <= 200:
charge = (100 * 2) + ((units - 100) * 3)
else:
charge = (100 * 2) + (100 * 3) + ((units - 200) * 5)
Add fixed charge
bill = charge + 50
Apply surcharge or rebate
if units > 300:
bill = bill + (bill * 0.10)
elif units < 100:
bill = bill - (bill * 0.05)
bills.append(bill)
Ln: 8 Col: 67
Rain warning
In effect
Search
ENG
IN
85%
12:14
28-10-2026


IDLE Shell 3.13.14
File Edit Shell Debug Options Window Help
Python 3.13.14 (tags/v3.13.14:fd17997, Jun 10 2026, 13:03:48) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
========== RESTART: C:/Users/Deepthi/OneDrive/Desktop/activity/case2.py ==========
Enter number of months: 2
Month 1
Enter electricity consumption (units): 2
Final Bill: Rs. 51.3
Month 2
Enter electricity consumption (units): 200
Final Bill: Rs. 550.0
--- MONTHLY BILL ANALYSIS ---
Month 1 : Rs. 51.3
Month 2 : Rs. 550.0
Status: Unusual increase
>>>
Ln: 44 Col: 35
Rain warning
In effect
Search
ENG
IN
86%
12:14
28-10-2026



INPUT number of months

previous_bill = none

FOR each month

INPUT units

CALCULATE slab charge

bill = slab charge + fixed charge

IF bill > surcharge threshold

ADD surcharge

ELSE IF units < rebate threshold

APPLY rebate

ENDIF

DISPLAY bill

IF previous_bill exists AND bill > 1.5 * previous_bill

DISPLAY unusual increase warning

```
  ENDIF
  previous_bill = bill
ENDFOR
```

Python Code

```
months = int(input("Number of months: "))
previous_bill = None

for month in range(1, months + 1):
  units = float(input(f"Units used in month {month}: "))

  if units < 0:
    print("Units cannot be negative.")
    continue

  if units <= 100:
    energy_charge = units * 2
  elif units <= 200:
    energy_charge = 100 * 2 + (units - 100) * 3
  else:
    energy_charge = 100 * 2 + 100 * 3 + (units - 200) * 5

  bill = energy_charge + 50

  if bill > 500:
    bill *= 1.05
  elif units < 100:
    bill *= 0.98
```

```
    print(f"Month {month} bill: {bill:.2f}")


    if previous_bill is not None and bill > previous_bill * 1.5:
        print("Warning: unusual increase from the previous month.")


    previous_bill = bill
```

Test Case and Expected Output

Input usage: 80 units, then 180 units.

Month 1 bill: 206.80

Month 2 bill: 390.00

No warning is displayed because 390 is not more than 1.5 times 206.80.

Conclusion

The program uses progressive slabs to calculate monthly bills, applies the stated adjustments, and identifies large month to month increases.

Case 3: Hospital Billing and Patient Classification

Problem Statement

Process consultation, diagnostic, and treatment charges; apply concessions; validate billing constraints; and classify patients according to payment status.

Objective

Calculate each patient's final bill, remaining balance, and payment classification.

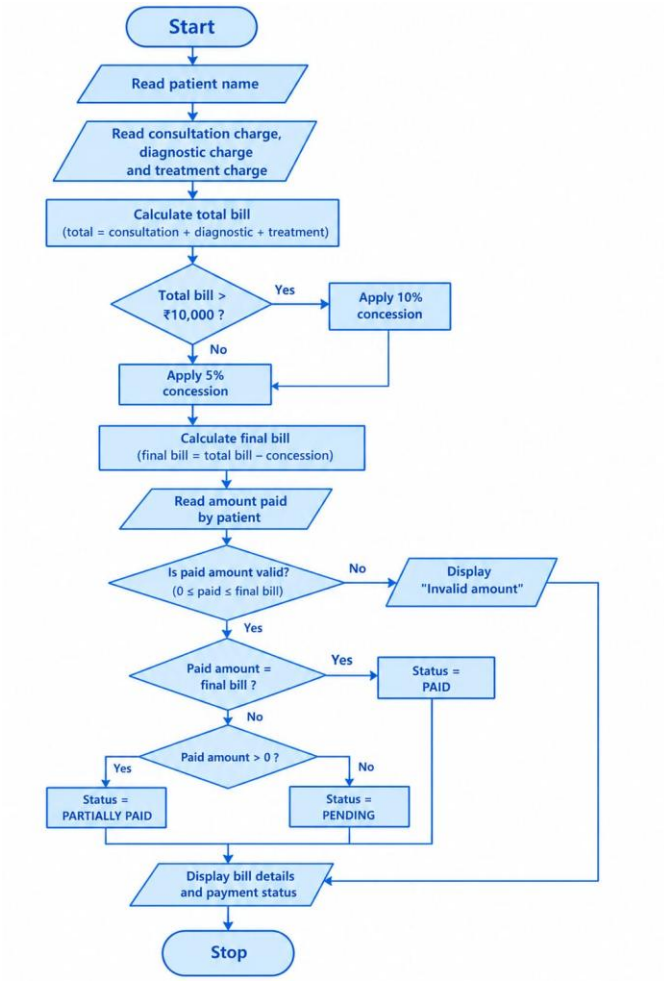
Assumptions

- Patients aged 60 or above receive a 10% concession.
- Otherwise, patients with monthly income below 20,000 receive a 15% concession.
- Only one concession is applied.
- Payment cannot be negative or exceed the final bill.
- Payment status is unpaid, partially paid, or paid.

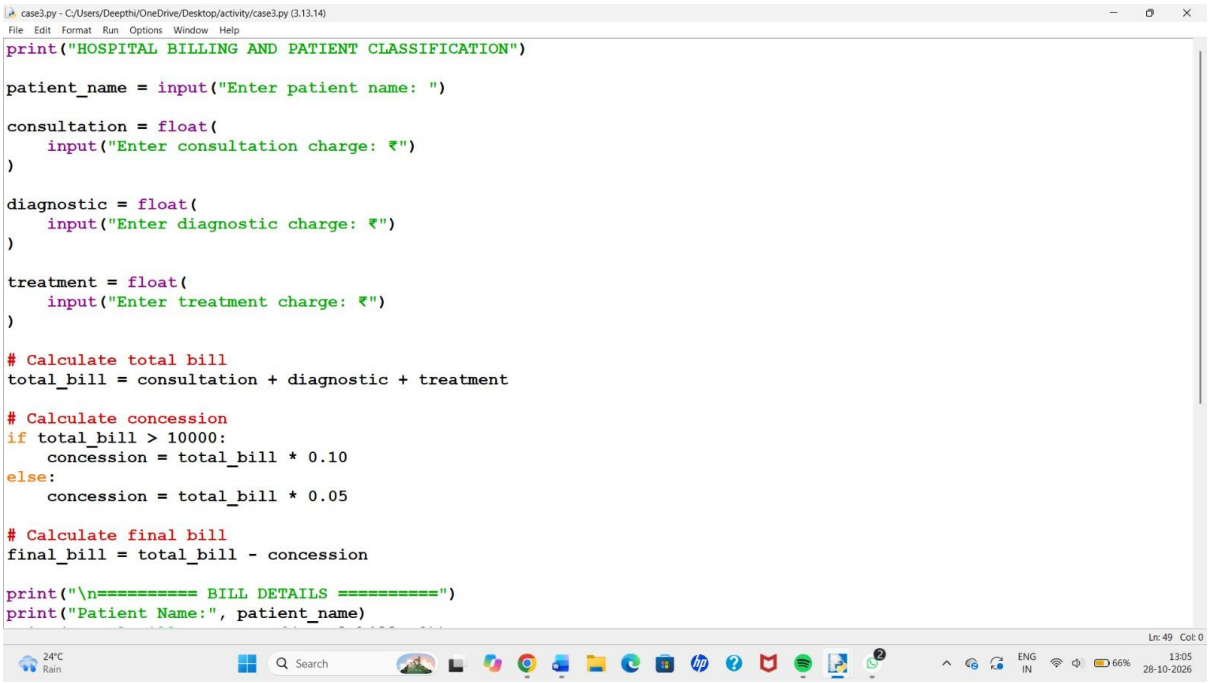
Algorithm

1. Read the number of patients.
2. For each patient, read identifying and billing details.
3. Validate that charges and payment are nonnegative.
4. Calculate gross charges and apply the applicable concession.
5. Validate payment against the final bill.
6. Calculate the balance and display patient category and payment status.

Flowchart



Pseudocode



```
print("HOSPITAL BILLING AND PATIENT CLASSIFICATION")

patient_name = input("Enter patient name: ")

consultation = float(
    input("Enter consultation charge: ₹")
)

diagnostic = float(
    input("Enter diagnostic charge: ₹")
)

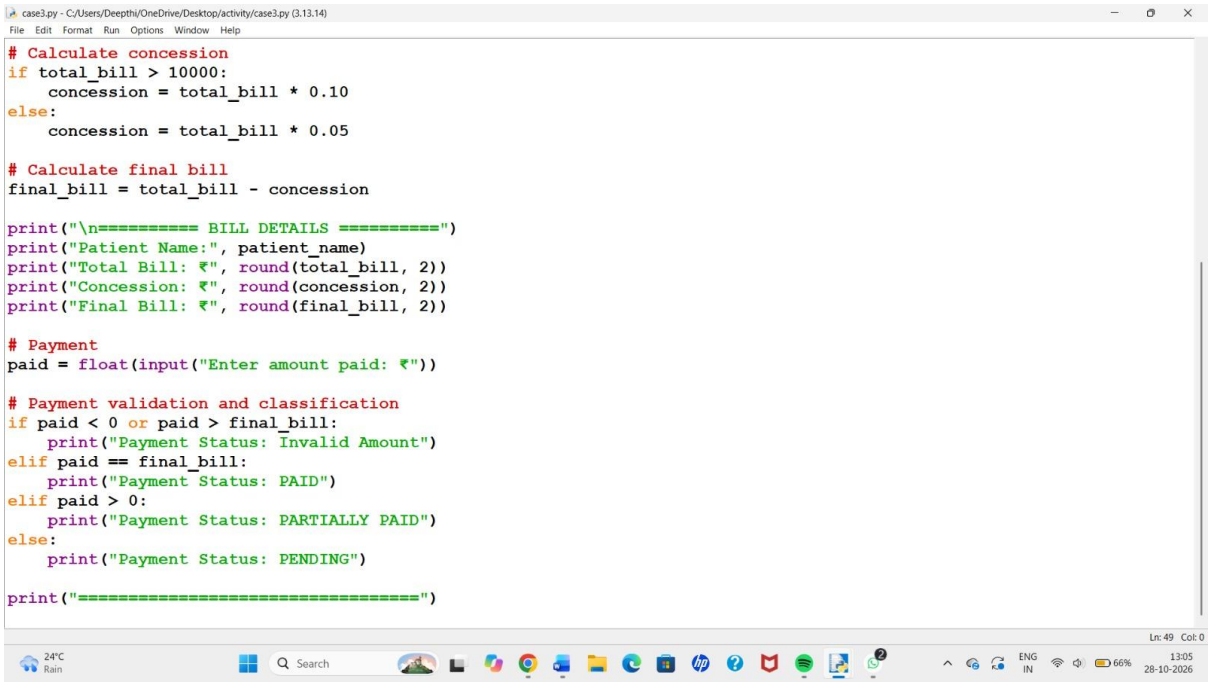
treatment = float(
    input("Enter treatment charge: ₹")
)

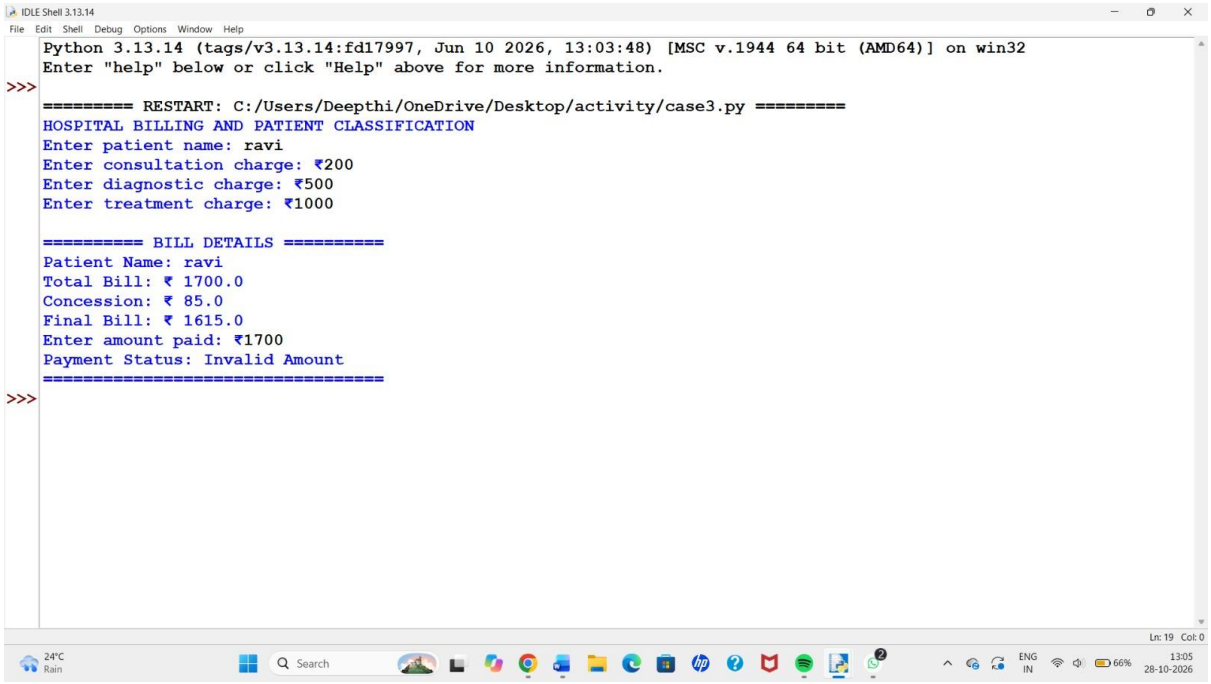
# Calculate total bill
total_bill = consultation + diagnostic + treatment

# Calculate concession
if total_bill > 10000:
    concession = total_bill * 0.10
else:
    concession = total_bill * 0.05

# Calculate final bill
final_bill = total_bill - concession

print("\n========== BILL DETAILS ==========")
print("Patient Name:", patient_name)
```





INPUT number of patients

FOR each patient

INPUT name, age, income, charges, payment

IF any charge or payment is negative

DISPLAY invalid input

ELSE

```
      total = sum of charges
      IF age >= 60
        concession = 10% of total
      ELSE IF income < 20000
        concession = 15% of total
      ELSE
        concession = 0
      ENDIF

      bill = total - concession
      IF payment > bill
        DISPLAY invalid payment
      ELSE
        balance = bill - payment
        CLASSIFY as paid, partially paid, or unpaid
        DISPLAY bill and classification
      ENDIF
    ENDIF
ENDFOR
```

Python Code

```
count = int(input("Number of patients: "))

for _ in range(count):
    print("\nPatient details")
    name = input("Name: ")
    age = int(input("Age: "))
    income = float(input("Monthly income: "))
    consultation = float(input("Consultation charge: "))
```

```
diagnostic = float(input("Diagnostic charge: "))
treatment = float(input("Treatment charge: "))
paid = float(input("Amount paid: "))

values = [income, consultation, diagnostic, treatment, paid]
if min(values) < 0:
    print("Invalid: charges, income, and payment cannot be negative.")
    continue

total = consultation + diagnostic + treatment

if age >= 60:
    concession_rate = 0.10
    category = "Senior"
elif income < 20000:
    concession_rate = 0.15
    category = "Concession"
else:
    concession_rate = 0
    category = "Regular"

bill = total * (1 - concession_rate)

if paid > bill:
    print("Invalid: payment cannot exceed the final bill.")
    continue

balance = bill - paid
```

```
    if balance == 0:
        status = "Paid"
    elif paid == 0:
        status = "Unpaid"
    else:
        status = "Partially paid"

    print(f"{name} | Category: {category} | Final bill: {bill:.2f}")
    print(f"Paid: {paid:.2f} | Balance: {balance:.2f} | Status: {status}")
```

Test Case and Expected Output

A patient aged 65 has charges of 500, 1,000, and 2,500, and pays 2,000. The 10% senior concession applies.

Ravi | Category: Senior | Final bill: 3600.00

Paid: 2000.00 | Balance: 1600.00 | Status: Partially paid

Conclusion

The program calculates patient bills, applies the selected concession, checks payment validity, and reports each patient's outstanding balance.

Case 4: Employee Payroll and Deduction Analysis

Problem Statement

Calculate gross salary from basic pay, allowances, overtime, and incentives. Apply progressive deductions and produce a formatted salary statement.

Objective

Calculate gross salary, income tax, other deductions, and take home salary.

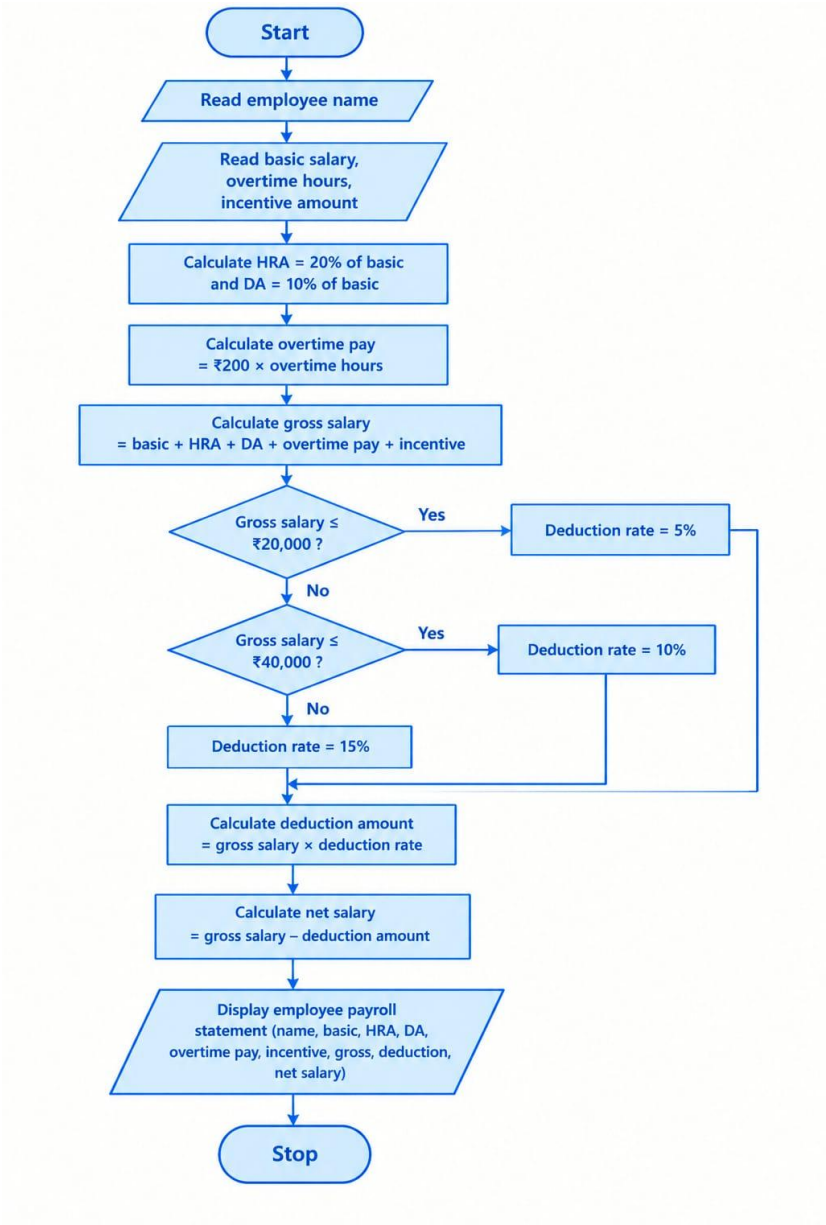
Assumptions

- Gross salary is basic pay plus allowances, overtime pay, and incentive.
- Tax is 0% on the first 20,000, 10% on the next 30,000, and 20% on income above 50,000.
- Other deductions are entered by the user.
- All entered values must be nonnegative.

Algorithm

1. Read employee salary details.
2. Calculate overtime pay and gross salary.
3. Calculate progressive tax according to the example brackets.
4. Add other deductions.
5. Calculate net salary and display the salary statement.

Flowchart


Start
Read employee name
Read basic salary, overtime hours, incentive amount
Calculate HRA = 20% of basic and DA = 10% of basic
Calculate overtime pay = ₹200 × overtime hours
Calculate gross salary = basic + HRA + DA + overtime pay + incentive
Gross salary ≤ ₹20,000 ?
Yes
Deduction rate = 5%
No
Gross salary ≤ ₹40,000 ?
Yes
Deduction rate = 10%
No
Deduction rate = 15%
Calculate deduction amount = gross salary × deduction rate
Calculate net salary = gross salary – deduction amount
Display employee payroll statement (name, basic, HRA, DA, overtime pay, incentive, gross, deduction, net salary)
Stop

Pseudocode


print("EMPLOYEE PAYROLL SYSTEM")
name = input("Enter employee name: ")
basic = float(input("Enter basic pay: "))
overtime_hours = float(input("Enter overtime hours: "))
incentive = float(input("Enter incentive amount: "))
Calculate allowances
hra = basic * 0.20
da = basic * 0.10
Calculate overtime pay
overtime_pay = overtime_hours * 200
Calculate gross salary
gross_salary = basic + hra + da + overtime_pay + incentive
Calculate progressive deduction
if gross_salary <= 20000:
deduction_rate = 0.05
elif gross_salary <= 40000:
deduction_rate = 0.10
else:
deduction_rate = 0.15
deduction = gross_salary * deduction_rate
Calculate net salary
net_salary = gross_salary - deduction





INPUT employee name, basic, allowances, overtime, rate, incentive, other deductions

gross = basic + allowances + overtime * rate + incentive

IF gross <= 20000

  tax = 0

ELSE IF gross <= 50000

  tax = 10% of amount above 20000

ELSE

  tax = 10% of 30000 + 20% of amount above 50000

ENDIF

net = gross - tax - other deductions

DISPLAY formatted salary statement

Python Code

```
name = input("Employee name: ")
basic = float(input("Basic pay: "))
allowances = float(input("Allowances: "))
overtime_hours = float(input("Overtime hours: "))
overtime_rate = float(input("Overtime rate per hour: "))
incentive = float(input("Incentive: "))
other_deductions = float(input("Other deductions: "))

inputs = [basic, allowances, overtime_hours, overtime_rate,
        incentive, other_deductions]

if min(inputs) < 0:
    print("Invalid: payroll amounts and hours cannot be negative.")
else:
    overtime_pay = overtime_hours * overtime_rate
    gross = basic + allowances + overtime_pay + incentive

    if gross <= 20000:
        tax = 0
    elif gross <= 50000:
        tax = (gross - 20000) * 0.10
    else:
        tax = 30000 * 0.10 + (gross - 50000) * 0.20

    total_deductions = tax + other_deductions
    net_salary = gross - total_deductions

    print(f"\nSalary Statement: {name}")
```

```
print(f"Basic pay:        {basic:.2f}")
print(f"Allowances:       {allowances:.2f}")
print(f"Overtime pay:     {overtime_pay:.2f}")
print(f"Incentive:        {incentive:.2f}")
print(f"Gross salary:     {gross:.2f}")
print(f"Income tax:       {tax:.2f}")
print(f"Other deductions:  {other_deductions:.2f}")
print(f"Net salary:       {net_salary:.2f}")
```

Test Case and Expected Output

Inputs: basic pay 40,000; allowances 5,000; 10 overtime hours at 200/hour; incentive 1,000; other deductions 1,000.

Gross salary: 48000.00

Income tax: 2800.00

Other deductions: 1000.00

Net salary: 44200.00

Conclusion

The program combines salary components, calculates progressive tax, and displays the employee's final net salary.

Case 5: Multi Plan Telecommunications Cost Optimization

Problem Statement

Evaluate mobile plans against usage, validity, and budget. Calculate excess usage charges and identify the most economical eligible plan.

Objective

Find the lowest cost plan that meets the required validity and budget.

Assumptions

- Each plan has a monthly fee, included minutes, included data, validity, and extra usage rates.
- Excess minutes and data are charged separately.
- A plan is eligible only if it meets the required validity and its calculated cost is within budget.
- Usage and costs cannot be negative.

Algorithm

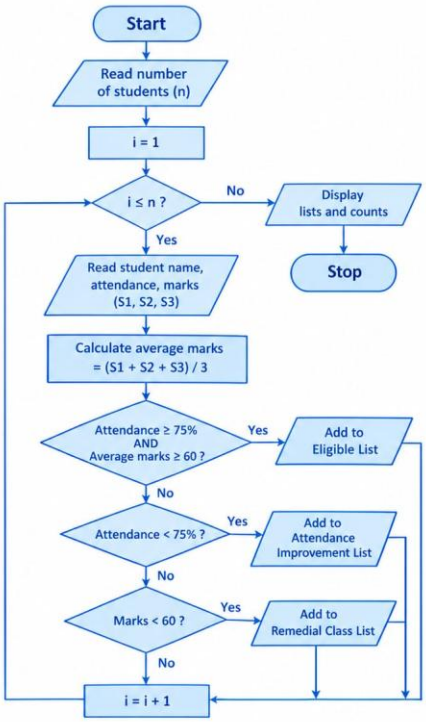
1. Read user requirements: validity, expected call minutes, expected data, and budget.
2. For each plan, read its fee, included usage, validity, and excess rates.
3. Ignore plans that do not meet the validity requirement.
4. Calculate each remaining plan's total cost, including excess usage.
5. Keep plans within budget and display the least expensive one.

Flowchart

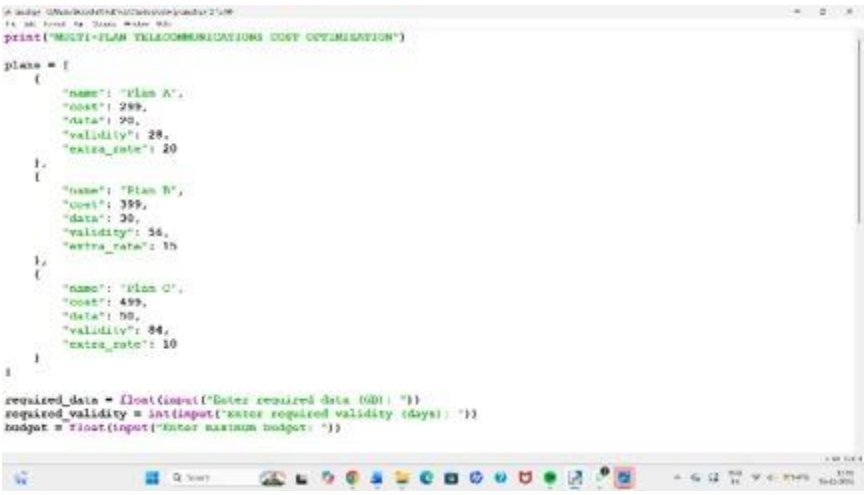
Case Study 5 – Student Attendance and Performance Tracking

Algorithm

1. Start
2. Read the number of students (n).
3. For each student (i = 1 to n) do:
   - 3.1 Read student name.
   - 3.2 Read attendance percentage.
   - 3.3 Read marks in Subject 1.
   - 3.4 Read marks in Subject 2.
   - 3.5 Read marks in Subject 3.
   - 3.6 Calculate average marks = (S1 + S2 + S3) / 3.
4. Check eligibility:
   - 4.1 Attendance ≥ 75%
   - 4.2 Average marks ≥ 60
5. If eligible, add the student to the eligible list.
6. If attendance < 75%, add the student to the attendance improvement list.
7. If marks < 60, add the student to the remedial class list.
8. Repeat for next student.
9. After all students are processed, display the number of students in each list and their details.
10. Stop



Pseudocode







INPUT requirements and number of plans

eligible_plans = empty list

FOR each plan

  INPUT plan fee, included usage, validity, excess rates

  IF plan validity >= required validity

    excess_cost = extra minutes cost + extra data cost

    total_cost = monthly fee + excess_cost

    IF total_cost <= budget

      ADD plan and total_cost to eligible_plans

    ENDIF

  ENDIF

ENDFOR

IF eligible_plans is empty

  DISPLAY no eligible plan

ELSE

  DISPLAY eligible plan with minimum total cost

ENDIF

Python Code

```
required_days = int(input("Required validity in days: "))
```

```
minutes_used = float(input("Expected call minutes: "))
data_used = float(input("Expected data usage in GB: "))
budget = float(input("Monthly budget: "))
count = int(input("Number of plans: "))

eligible = []

for i in range(count):
    print(f"\nPlan {i + 1}")
    name = input("Plan name: ")
    fee = float(input("Monthly fee: "))
    included_minutes = float(input("Included call minutes: "))
    included_data = float(input("Included data in GB: "))
    validity = int(input("Validity in days: "))
    extra_minute_rate = float(input("Cost per extra minute: "))
    extra_gb_rate = float(input("Cost per extra GB: "))

    values = [fee, included_minutes, included_data,
              extra_minute_rate, extra_gb_rate]

    if min(values) < 0 or validity < 0:
        print("Invalid plan values; plan ignored.")
        continue

    if validity >= required_days:
        extra_minutes = max(0, minutes_used - included_minutes)
        extra_data = max(0, data_used - included_data)
        total_cost = fee + extra_minutes * extra_minute_rate
```

```
        total_cost += extra_data * extra_gb_rate

        if total_cost <= budget:
            eligible.append([name, total_cost])

if eligible:
    best = min(eligible, key=lambda plan: plan[1])
    print(f"\nMost economical eligible plan: {best[0]}")
    print(f"Estimated monthly cost: {best[1]:.2f}")
else:
    print("\nNo plan meets both validity and budget requirements.")
```

Test Case and Expected Output

Requirements: 30 days, 200 call minutes, 5 GB, budget 500. Plan Alpha costs 300, includes 250 minutes and 6 GB, and is valid for 30 days. Plan Beta costs 250, includes 100 minutes and 3 GB, charges 1 per extra minute and 50 per extra GB, and is valid for 30 days.

Most economical eligible plan: Alpha

Estimated monthly cost: 300.00

Conclusion

The program estimates the full cost of each qualifying plan, including excess usage, and selects the lowest cost plan within the budget.

Case 6: Railway Fare and Reservation Calculator

Problem Statement

Calculate fares for multiple passengers using distance slabs, travel class, age concessions, reservation fees, and cancellation penalties.

Objective

Calculate each passenger's fare, total reservation cost, and refund after cancellation.

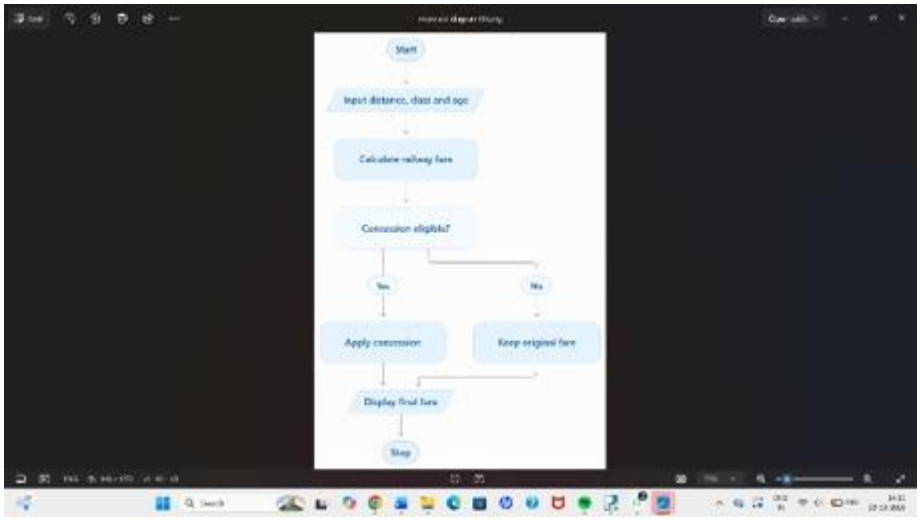
Assumptions

- Sleeper fare: 1 per kilometre; AC fare: 2 per kilometre.
- Children under 5 travel free but pay the reservation fee.
- Passengers aged 60 or over receive a 20% fare concession.
- Reservation fee: 30 per passenger.
- Cancellation penalty: 20% of the total fare, including reservation fees.

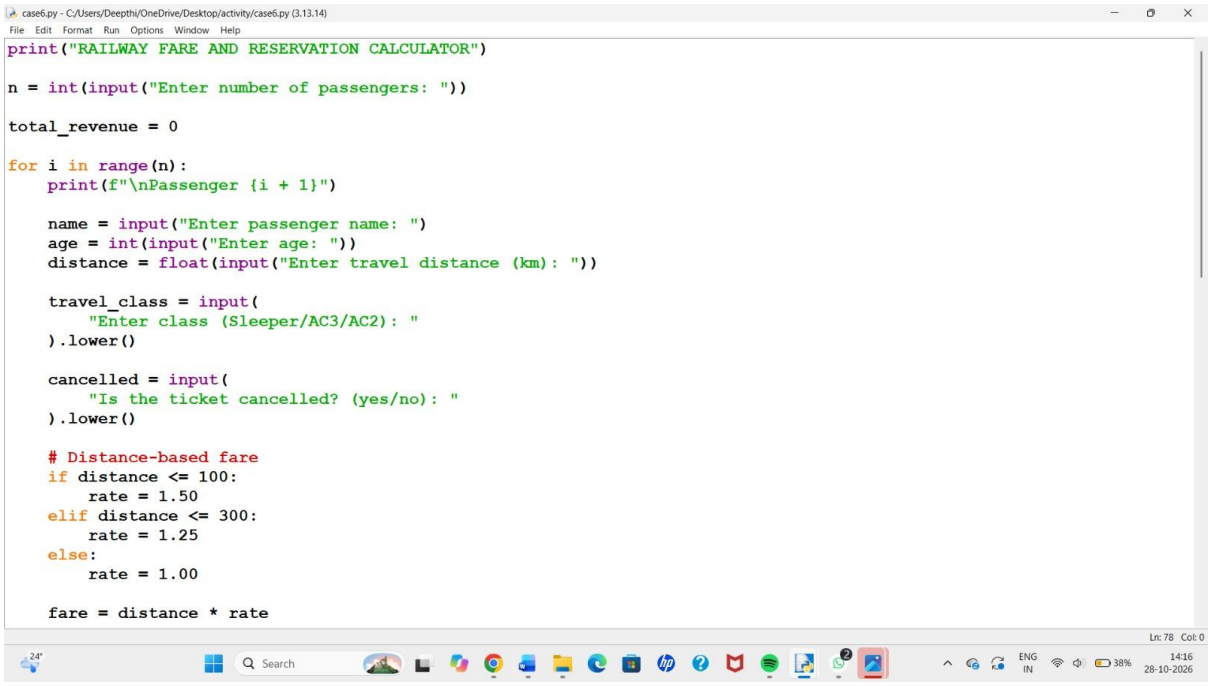
Algorithm

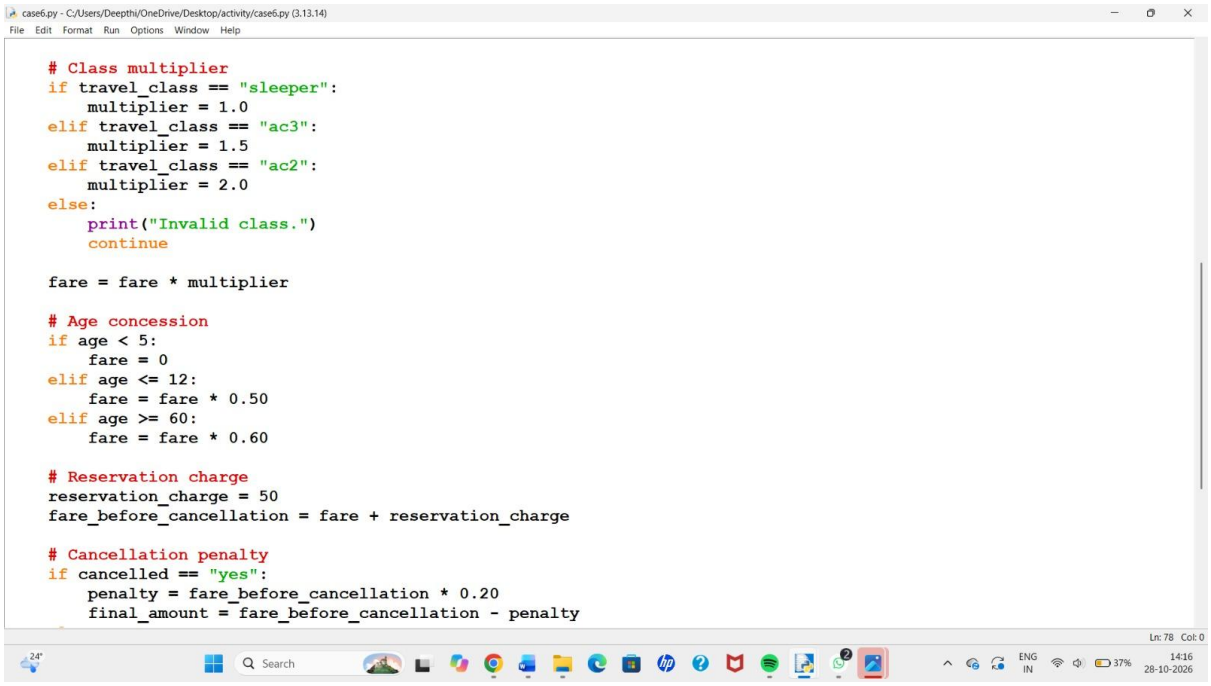
1. Read journey distance, travel class, and passenger count.
2. Calculate the fare rate for the selected class.
3. For each passenger, calculate fare after any age concession.
4. Add the reservation fee and sum the total.
5. If the user cancels, calculate the penalty and refund.

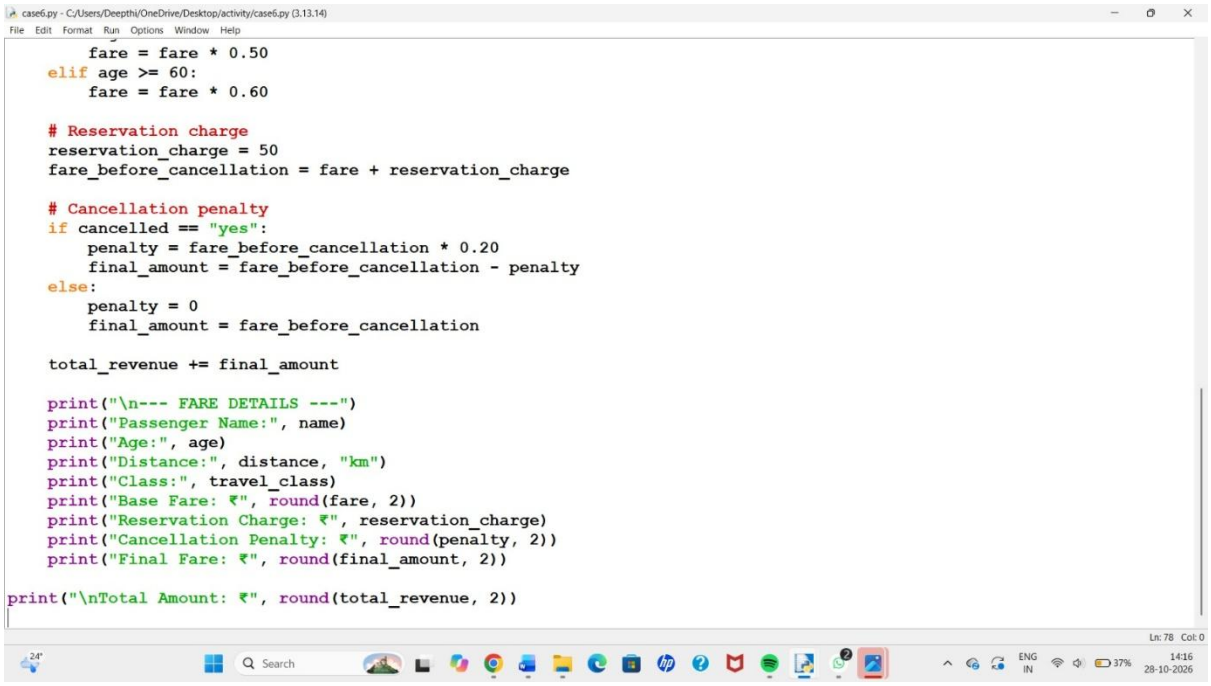
Flowchart



Pseudocode


case6.py - C:/Users/Deepthi/OneDrive/Desktop/activity/case6.py (3.13.14)
File Edit Format Run Options Window Help
print("RAILWAY FARE AND RESERVATION CALCULATOR")
n = int(input("Enter number of passengers: "))
total_revenue = 0
for i in range(n):
print(f"\nPassenger {i + 1}")
name = input("Enter passenger name: ")
age = int(input("Enter age: "))
distance = float(input("Enter travel distance (km): "))
travel_class = input(
"Enter class (Sleeper/AC3/AC2): "
).lower()
cancelled = input(
"Is the ticket cancelled? (yes/no): "
).lower()
Distance-based fare
if distance <= 100:
rate = 1.50
elif distance <= 300:
rate = 1.25
else:
rate = 1.00
fare = distance * rate
Ln: 78 Col: 0
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28-10-2026


case6.py - C:/Users/Deepthi/OneDrive/Desktop/activity/case6.py (3.13.14)
File Edit Format Run Options Window Help
Class multiplier
if travel_class == "sleeper":
multiplier = 1.0
elif travel_class == "ac3":
multiplier = 1.5
elif travel_class == "ac2":
multiplier = 2.0
else:
print("Invalid class.")
continue
fare = fare * multiplier
Age concession
if age < 5:
fare = 0
elif age <= 12:
fare = fare * 0.50
elif age >= 60:
fare = fare * 0.60
Reservation charge
reservation_charge = 50
fare_before_cancellation = fare + reservation_charge
Cancellation penalty
if cancelled == "yes":
penalty = fare_before_cancellation * 0.20
final_amount = fare_before_cancellation - penalty
Ln: 78 Col: 0
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28-10-2026



```
        fare = fare * 0.50
    elif age >= 60:
        fare = fare * 0.60

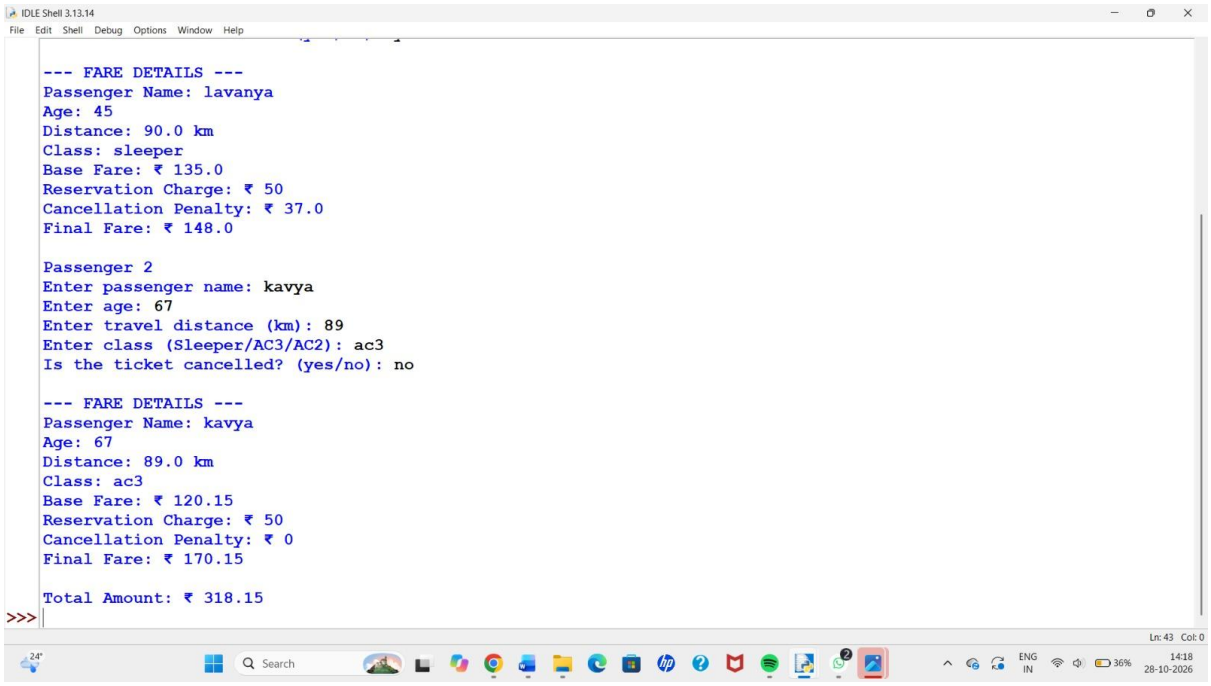
    # Reservation charge
    reservation_charge = 50
    fare_before_cancellation = fare + reservation_charge

    # Cancellation penalty
    if cancelled == "yes":
        penalty = fare_before_cancellation * 0.20
        final_amount = fare_before_cancellation - penalty
    else:
        penalty = 0
        final_amount = fare_before_cancellation

    total_revenue += final_amount

    print("\n--- FARE DETAILS ---")
    print("Passenger Name:", name)
    print("Age:", age)
    print("Distance:", distance, "km")
    print("Class:", travel_class)
    print("Base Fare: ₹", round(fare, 2))
    print("Reservation Charge: ₹", reservation_charge)
    print("Cancellation Penalty: ₹", round(penalty, 2))
    print("Final Fare: ₹", round(final_amount, 2))

print("\nTotal Amount: ₹", round(total_revenue, 2))
```



```
--- FARE DETAILS ---
Passenger Name: lavanya
Age: 45
Distance: 90.0 km
Class: sleeper
Base Fare: ₹ 135.0
Reservation Charge: ₹ 50
Cancellation Penalty: ₹ 37.0
Final Fare: ₹ 148.0

Passenger 2
Enter passenger name: kavya
Enter age: 67
Enter travel distance (km): 89
Enter class (Sleeper/AC3/AC2): ac3
Is the ticket cancelled? (yes/no): no

--- FARE DETAILS ---
Passenger Name: kavya
Age: 67
Distance: 89.0 km
Class: ac3
Base Fare: ₹ 120.15
Reservation Charge: ₹ 50
Cancellation Penalty: ₹ 0
Final Fare: ₹ 170.15

Total Amount: ₹ 318.15
>>>
```

INPUT distance, class, passenger count

SET rate based on class

total = 0

FOR each passenger

  INPUT age

  fare = distance * rate

```
  IF age < 5
    fare = 0
  ELSE IF age >= 60
    fare = fare * 0.80
  ENDIF
  total = total + fare + reservation fee
ENDFOR

DISPLAY total
IF cancelled
  penalty = total * cancellation rate
  DISPLAY refund = total - penalty
ENDIF
```

Python Code

```
distance = float(input("Journey distance in km: "))
travel_class = input("Class (sleeper/ac): ").lower()
passenger_count = int(input("Number of passengers: "))

if distance < 0 or passenger_count <= 0:
  print("Invalid distance or passenger count.")
else:
  if travel_class == "sleeper":
    rate = 1.0
  elif travel_class == "ac":
    rate = 2.0
  else:
    rate = None
```

```
    if rate is None:
        print("Invalid class.")
    else:
        reservation_fee = 30
        total = 0

        for i in range(passenger_count):
            age = int(input(f"Passenger {i + 1} age: "))
            fare = distance * rate

            if age < 5:
                fare = 0
            elif age >= 60:
                fare *= 0.80

            passenger_total = fare + reservation_fee
            total += passenger_total
            print(f"Passenger {i + 1} fare: {passenger_total:.2f}")

        print(f"Total reservation cost: {total:.2f}")

        cancel = input("Cancel reservation? (yes/no): ").lower()
        if cancel == "yes":
            penalty = total * 0.20
            print(f"Cancellation penalty: {penalty:.2f}")
            print(f"Refund: {total - penalty:.2f}")
```

Test Case and Expected Output

Distance: 100 km; class: sleeper; two passengers aged 30 and 65. First passenger pays 100 + 30. Second pays 80 + 30.

Passenger 1 fare: 130.00

Passenger 2 fare: 110.00

Total reservation cost: 240.00

Cancel reservation? (yes/no): yes

Cancellation penalty: 48.00

Refund: 192.00

Conclusion

The program calculates passenger fares based on distance, travel class, age concessions, reservation charges, and cancellation rules.

Case 7: Retail Billing with Multiple Discount Rules

Problem Statement

Process multiple products and quantities, apply category-specific taxes and threshold discounts, and generate an itemized invoice with a maximum concession limit.

Objective

Calculate each product's taxable amount and tax, apply a total discount subject to a cap, and print the final invoice.

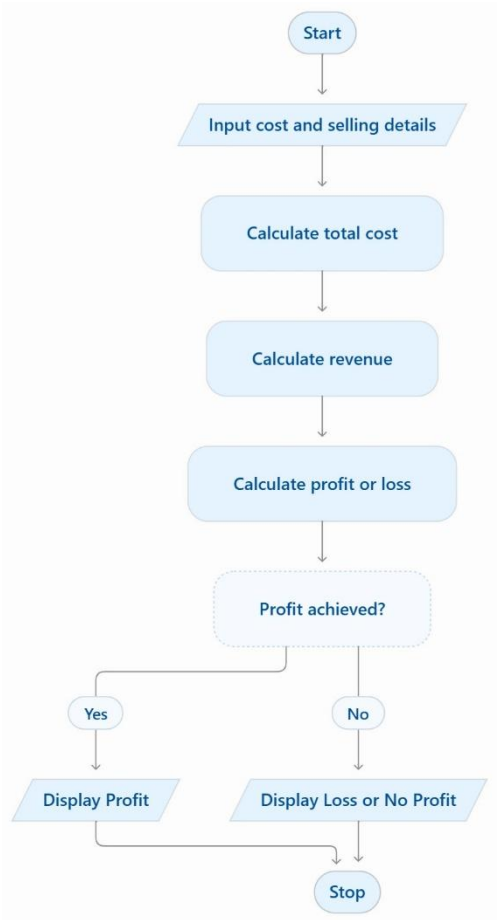
Assumptions

- Each product has a user-entered price, quantity, and tax rate.
- A 5% discount applies to the subtotal when it reaches the entered threshold.
- The total discount cannot exceed the entered maximum.
- Discount is distributed proportionally across items before tax.

Algorithm

1. Read each product's details and calculate line totals.
2. Add line totals to get the subtotal.
3. Check whether the subtotal reaches the discount threshold.
4. Calculate the discount and limit it to the maximum amount.
5. Allocate the discount proportionally to each product.
6. Calculate tax for each item and display the invoice totals.

Flowchart



Pseudocode

casee7.py - C:/Users/Deepthi/OneDrive/Desktop/activity/casee7.py (3.13.14)

File Edit Format Run Options Window Help

```
print("RETAIL BILLING SYSTEM")

products = []

n = int(input("Enter number of products: "))

for i in range(n):
    print(f"\nProduct {i + 1}")

    name = input("Enter product name: ")
    category = input(
        "Enter category (Grocery/Electronics/Clothing): "
    ).lower()
    quantity = int(input("Enter quantity: "))
    price = float(input("Enter unit price: "))

    subtotal = quantity * price

    products.append({
        "name": name,
        "category": category,
        "quantity": quantity,
        "price": price,
        "subtotal": subtotal
    })

total_subtotal = sum(item["subtotal"] for item in products)

# Discount calculation
if total_subtotal > 5000:
```

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```
    discount = total_subtotal * 0.10
elif total_subtotal >= 2500:
    discount = total_subtotal * 0.05
else:
    discount = 0

# Maximum concession limit
if discount > 1000:
    discount = 1000

discounted_total = total_subtotal - discount

# Tax calculation
total_tax = 0

for item in products:

    if item["category"] == "grocery":
        tax_rate = 0.05
    elif item["category"] == "electronics":
        tax_rate = 0.18
    elif item["category"] == "clothing":
        tax_rate = 0.12
    else:
        tax_rate = 0

    item["tax"] = item["subtotal"] * tax_rate
    total_tax += item["tax"]

final_amount = discounted_total + total_tax
```

INPUT product count

```
Python 3.13.14 (tags/v3.13.14:fd17997, Jun 10 2026, 13:03:48) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
========= RESTART: C:/Users/Deepthi/OneDrive/Desktop/activity/casee7.py ========
RETAIL BILLING SYSTEM
Enter number of products: 2

Product 1
Enter product name: refregirator
Enter category (Grocery/Electronics/Clothing): electronics
Enter quantity: 1
Enter unit price: 50000

Product 2
Enter product name: redgram
Enter category (Grocery/Electronics/Clothing): grocery
Enter quantity: 2
Enter unit price: 40

========== ITEMIZED INVOICE ==========
refregirator | Qty: 1 | Price: ₹50000.00 | Subtotal: ₹50000.00 | Tax: ₹9000.00
redgram | Qty: 2 | Price: ₹40.00 | Subtotal: ₹80.00 | Tax: ₹4.00
---------------------------------------
Total Subtotal: ₹50080.00
Discount: ₹1000.00
Tax: ₹9004.00
Final Amount: ₹58084.00
=======================================
>>>
>>>
```

subtotal = 0

FOR each product

INPUT name, quantity, price, tax rate

line total = quantity * price

```
    STORE product details
    subtotal = subtotal + line total
ENDFOR

INPUT discount threshold and maximum discount
IF subtotal >= threshold
    discount = minimum(5% of subtotal, maximum discount)
ELSE
    discount = 0
ENDIF

FOR each product
    allocate discount proportionally
    calculate tax on discounted line total
    DISPLAY product, net amount, and tax
ENDFOR

DISPLAY subtotal, discount, tax total, and final total
```

Python Code

```
products = []
subtotal = 0

count = int(input("Number of products: "))

for i in range(count):
    print(f"\nProduct {i + 1}")
    name = input("Product name: ")
    quantity = int(input("Quantity: "))
```

```
    price = float(input("Unit price: "))
    tax_rate = float(input("Category tax rate (%): "))

    if quantity < 0 or price < 0 or tax_rate < 0:
        print("Invalid values; enter nonnegative quantity, price, and tax.")
        raise SystemExit

    line_total = quantity * price
    subtotal += line_total
    products.append([name, quantity, line_total, tax_rate])

threshold = float(input("\nDiscount threshold: "))
max_discount = float(input("Maximum discount: "))

if subtotal >= threshold:
    discount = min(subtotal * 0.05, max_discount)
else:
    discount = 0

tax_total = 0
print("\nItemized Invoice")

for name, quantity, line_total, tax_rate in products:
    item_discount = discount * line_total / subtotal if subtotal else 0
    taxable_amount = line_total - item_discount
    tax = taxable_amount * tax_rate / 100
    tax_total += tax
```

```
    print(
        f"{name}: quantity={quantity}, "
        f"net={taxable_amount:.2f}, tax={tax:.2f}"
    )

final_total = subtotal - discount + tax_total
print(f"\nSubtotal: {subtotal:.2f}")
print(f"Discount: {discount:.2f}")
print(f"Tax total: {tax_total:.2f}")
print(f"Final total: {final_total:.2f}")
```

Test Case and Expected Output

Products: two notebooks at 100 each, taxed at 5%; one bag at 500, taxed at 10%.
Discount threshold: 500; maximum discount: 100.

Itemized Invoice

Notebook: quantity=2, net=190.00, tax=9.50

Bag: quantity=1, net=475.00, tax=47.50

Subtotal: 700.00

Discount: 35.00

Tax total: 57.00

Final total: 722.00

Conclusion

The program calculates an itemized bill with proportional discounts, category-specific taxes, and a maximum discount limit.

Case 8: Manufacturing Cost and Profitability Analysis

Problem Statement

Calculate material, labour, wastage, and machine costs; determine selling revenue and whether production meets a specified profit target.

Objective

Calculate total production cost, revenue, profit, and whether the profit target is met.

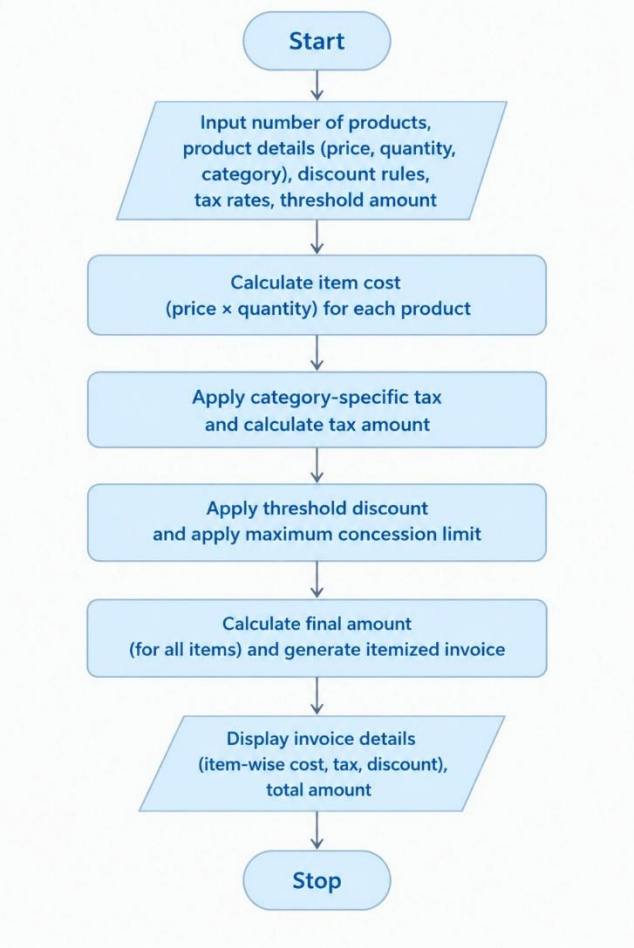
Assumptions

- Material cost is adjusted for the entered wastage percentage.
- Labour cost equals labour hours multiplied by the hourly rate.
- Machine cost equals machine hours multiplied by the machine rate.
- Revenue equals the number of units multiplied by the selling price per unit.

Algorithm

1. Read production quantity and cost details.
2. Add material wastage to the required material cost.
3. Calculate labour and machine costs.
4. Add all production costs.
5. Calculate revenue and profit.
6. Compare profit with the target and display the result.

Flowchart



Pseudocode

case8.py - C:/Users/Deepthi/OneDrive/Desktop/activity/case8.py (3.13.14)

File Edit Format Run Options Window Help

```
print("MANUFACTURING COST AND PROFITABILITY ANALYSIS")

quantity = int(input("Enter quantity produced: "))

material_per_unit = float(
    input("Enter material cost per unit: ₹")
)

labour_per_unit = float(
    input("Enter labour cost per unit: ₹")
)

machine_per_unit = float(
    input("Enter machine cost per unit: ₹")
)

wastage_percentage = float(
    input("Enter wastage percentage: ")
)

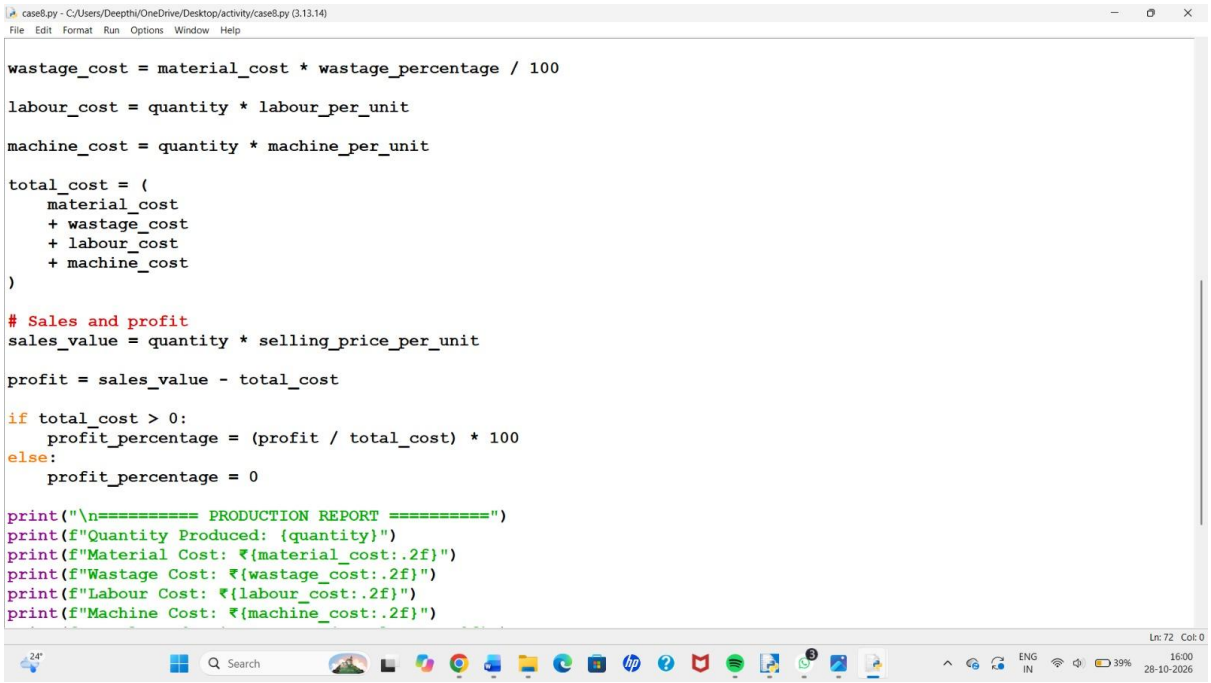
selling_price_per_unit = float(
    input("Enter selling price per unit: ₹")
)

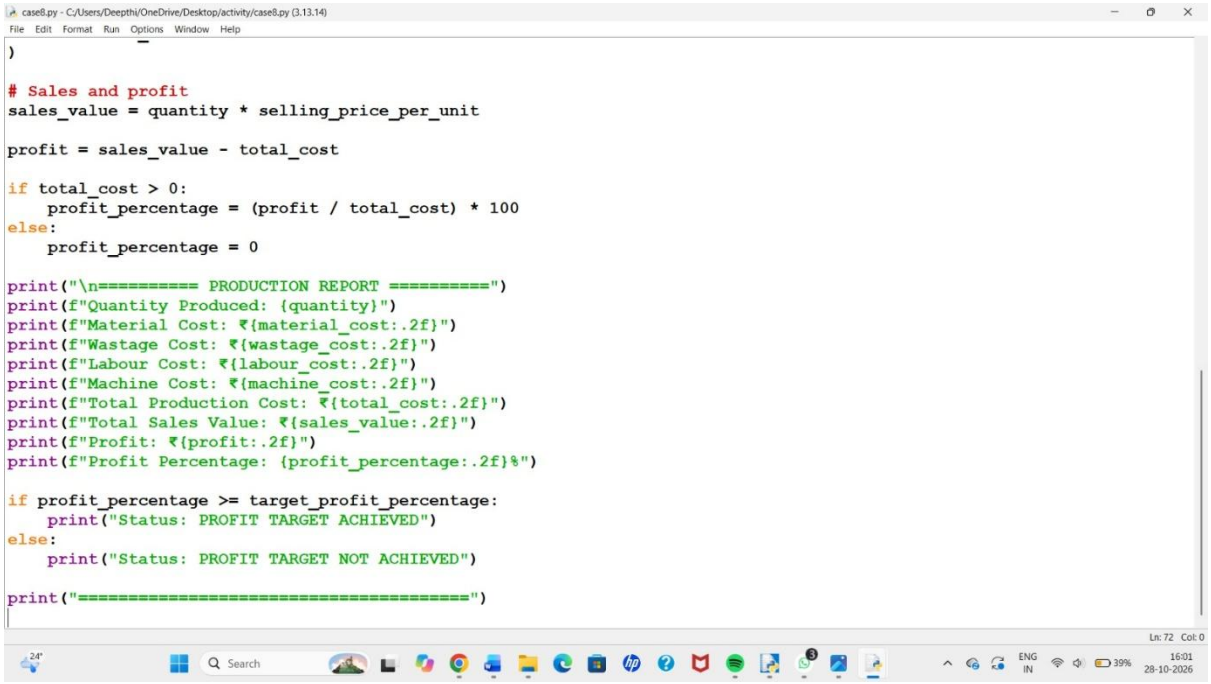
target_profit_percentage = float(
    input("Enter target profit percentage: ")
)

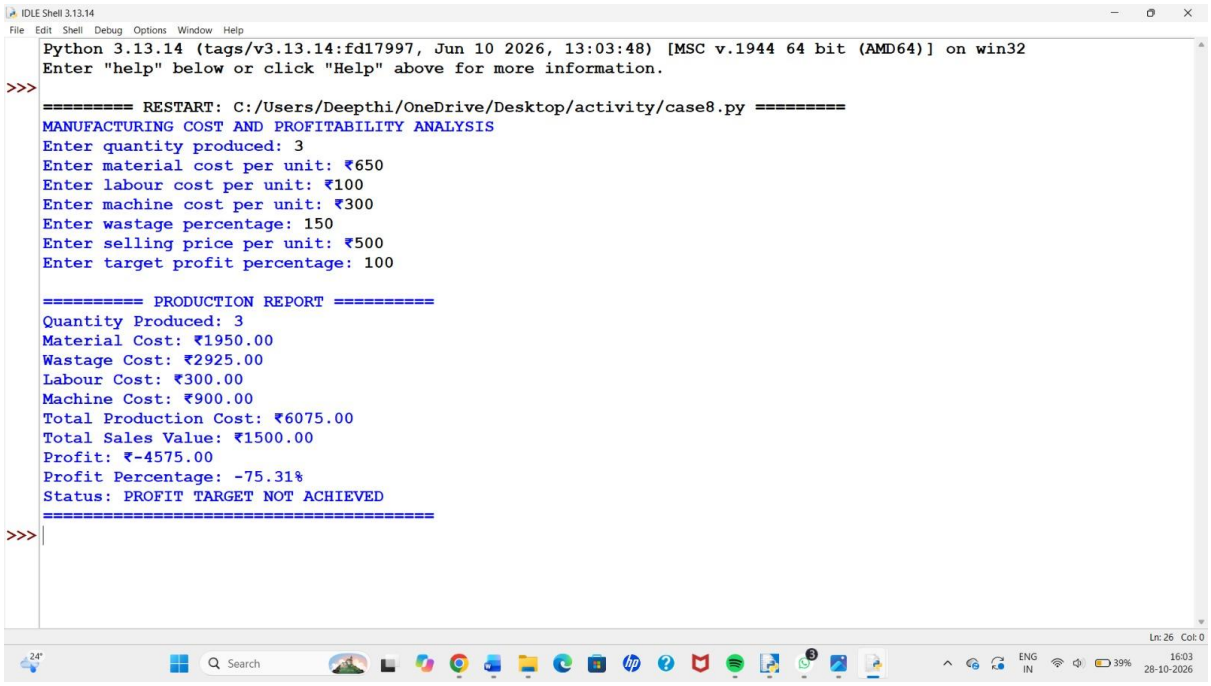
# Cost calculations
material_cost = quantity * material_per_unit
```

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case8.py - C:/Users/Deepthi/OneDrive/Desktop/activity/case8.py (3.13.14)
File Edit Format Run Options Window Help
wastage_cost = material_cost * wastage_percentage / 100
labour_cost = quantity * labour_per_unit
machine_cost = quantity * machine_per_unit
total_cost = (
material_cost
+ wastage_cost
+ labour_cost
+ machine_cost
)
Sales and profit
sales_value = quantity * selling_price_per_unit
profit = sales_value - total_cost
if total_cost > 0:
profit_percentage = (profit / total_cost) * 100
else:
profit_percentage = 0
print("\n========== PRODUCTION REPORT ==========")
print(f"Quantity Produced: {quantity}")
print(f"Material Cost: ₹{material_cost:.2f}")
print(f"Wastage Cost: ₹{wastage_cost:.2f}")
print(f"Labour Cost: ₹{labour_cost:.2f}")
print(f"Machine Cost: ₹{machine_cost:.2f}")
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case8.py - C:/Users/Deepthi/OneDrive/Desktop/activity/case8.py (3.13.14)
File Edit Format Run Options Window Help
)
Sales and profit
sales_value = quantity * selling_price_per_unit
profit = sales_value - total_cost
if total_cost > 0:
profit_percentage = (profit / total_cost) * 100
else:
profit_percentage = 0
print("\n========== PRODUCTION REPORT ==========")
print(f"Quantity Produced: {quantity}")
print(f"Material Cost: ₹{material_cost:.2f}")
print(f"Wastage Cost: ₹{wastage_cost:.2f}")
print(f"Labour Cost: ₹{labour_cost:.2f}")
print(f"Machine Cost: ₹{machine_cost:.2f}")
print(f"Total Production Cost: ₹{total_cost:.2f}")
print(f"Total Sales Value: ₹{sales_value:.2f}")
print(f"Profit: ₹{profit:.2f}")
print(f"Profit Percentage: {profit_percentage:.2f}%")
if profit_percentage >= target_profit_percentage:
print("Status: PROFIT TARGET ACHIEVED")
else:
print("Status: PROFIT TARGET NOT ACHIEVED")
print("==")
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INPUT units, material quantity and rate, wastage percentage

INPUT labour hours and rate, machine hours and rate

INPUT selling price per unit and profit target

material cost = quantity * rate * (1 + wastage / 100)

labour cost = labour hours * labour rate

machine cost = machine hours * machine rate

total cost = material cost + labour cost + machine cost

revenue = units * selling price

profit = revenue - total cost

IF profit >= target

  DISPLAY target achieved

ELSE

  DISPLAY target not achieved

ENDIF

Python Code

```
units = int(input("Number of units to produce: "))
material_quantity = float(input("Material quantity required: "))
material_rate = float(input("Material cost per unit: "))
wastage_percent = float(input("Wastage percentage: "))
labour_hours = float(input("Labour hours: "))
labour_rate = float(input("Labour cost per hour: "))
machine_hours = float(input("Machine hours: "))
machine_rate = float(input("Machine cost per hour: "))
selling_price = float(input("Selling price per finished unit: "))
profit_target = float(input("Profit target: "))

values = [
    units, material_quantity, material_rate, wastage_percent,
    labour_hours, labour_rate, machine_hours, machine_rate,
    selling_price, profit_target
]

if min(values) < 0:
    print("Invalid: enter nonnegative values.")
else:
    material_cost = (
        material_quantity * material_rate * (1 + wastage_percent / 100)
    )
    labour_cost = labour_hours * labour_rate
    machine_cost = machine_hours * machine_rate
    total_cost = material_cost + labour_cost + machine_cost
    revenue = units * selling_price
    profit = revenue - total_cost
```

```
print(f"\nMaterial cost: {material_cost:.2f}")
print(f"Labour cost: {labour_cost:.2f}")
print(f"Machine cost: {machine_cost:.2f}")
print(f"Total production cost: {total_cost:.2f}")
print(f"Revenue: {revenue:.2f}")
print(f"Profit: {profit:.2f}")

if profit >= profit_target:
    print("Production meets the profit target.")
else:
    print("Production does not meet the profit target.")
```

Test Case and Expected Output

Produce 100 units. Material: 100 units at 10 each, with 5% wastage. Labour: 20 hours at 100/hour. Machine: 10 hours at 50/hour. Selling price: 20/unit. Target profit: 500.

Material cost: 1050.00

Labour cost: 2000.00

Machine cost: 500.00

Total production cost: 3550.00

Revenue: 2000.00

Profit: -1550.00

Production does not meet the profit target.

Conclusion

The program totals the stated production costs, compares them with expected sales revenue, and checks the result against the profit target.

Case 9: Multi Currency Travel Budget Analysis

Problem Statement

Convert travel expenses using supplied exchange rates, include conversion fees, compare planned spending with a budget, and identify the largest expense category.

Objective

Calculate total travel spending in the home currency, determine whether spending is within budget, and find the largest expense category.

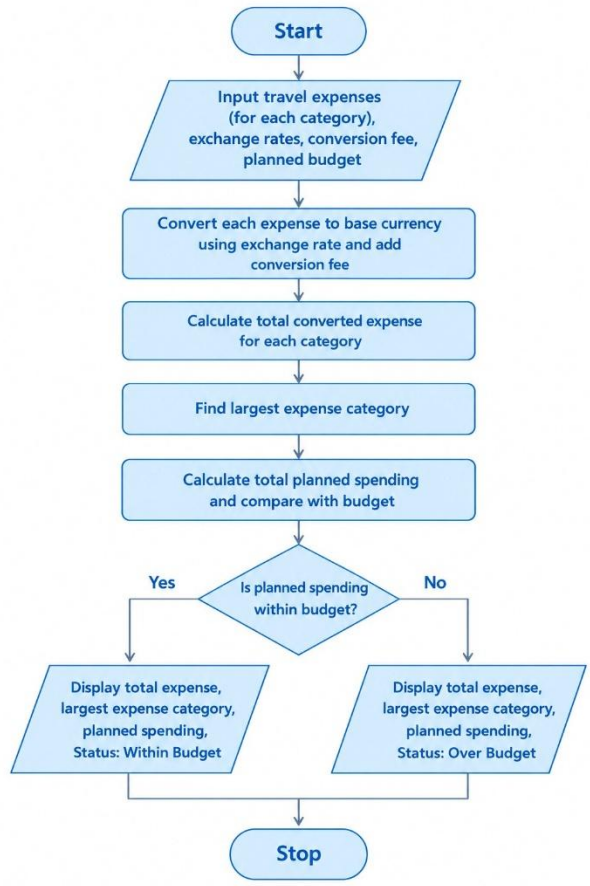
Assumptions

- The exchange rate is entered as home currency per one unit of foreign currency.
- Conversion fee is a percentage of the converted amount.
- Expense amounts and fees cannot be negative.
- The largest category is based on the final converted amount including the fee.

Algorithm

1. Read each expense category, foreign amount, exchange rate, and conversion fee.
2. Convert the expense to the home currency.
3. Add the conversion fee.
4. Sum all converted expenses.
5. Compare the total with the budget.
6. Display the largest expense category.

Flowchart



Pseudocode

```
print("MULTI-CURRENCY TRAVEL BUDGET ANALYSIS")

budget = float(input("Enter travel budget in INR: ₹"))

conversion_fee_percentage = float(
    input("Enter conversion fee percentage: ")
)

n = int(input("Enter number of expenses: "))

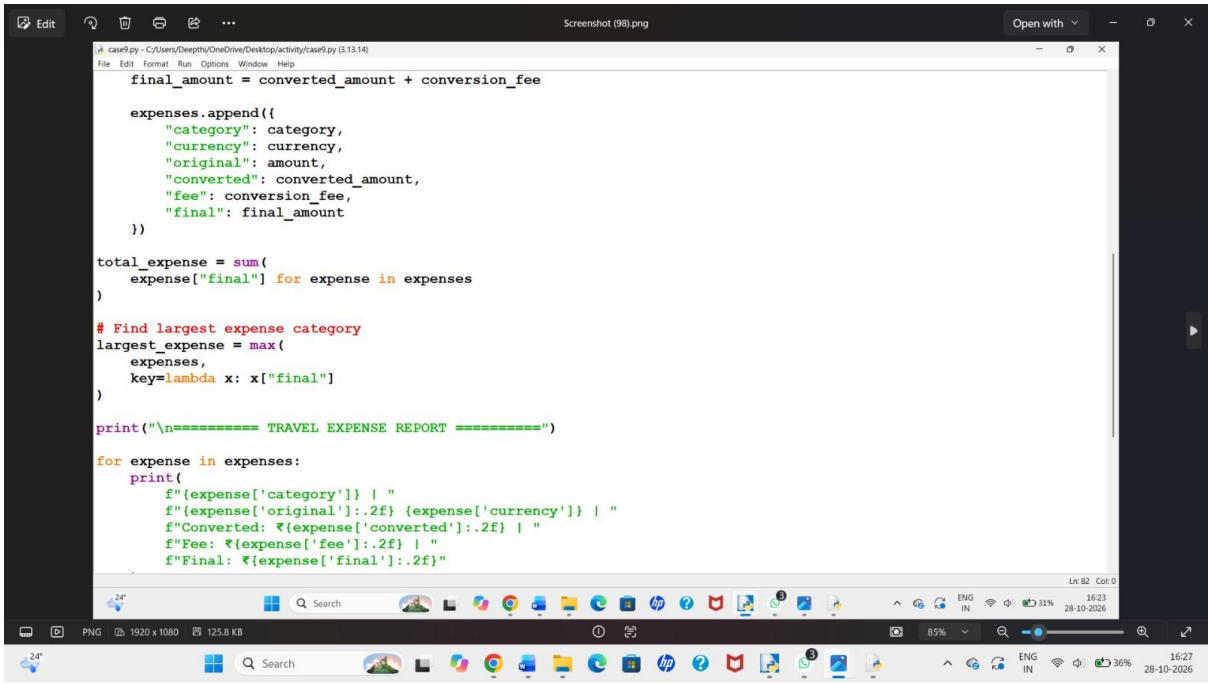
expenses = []

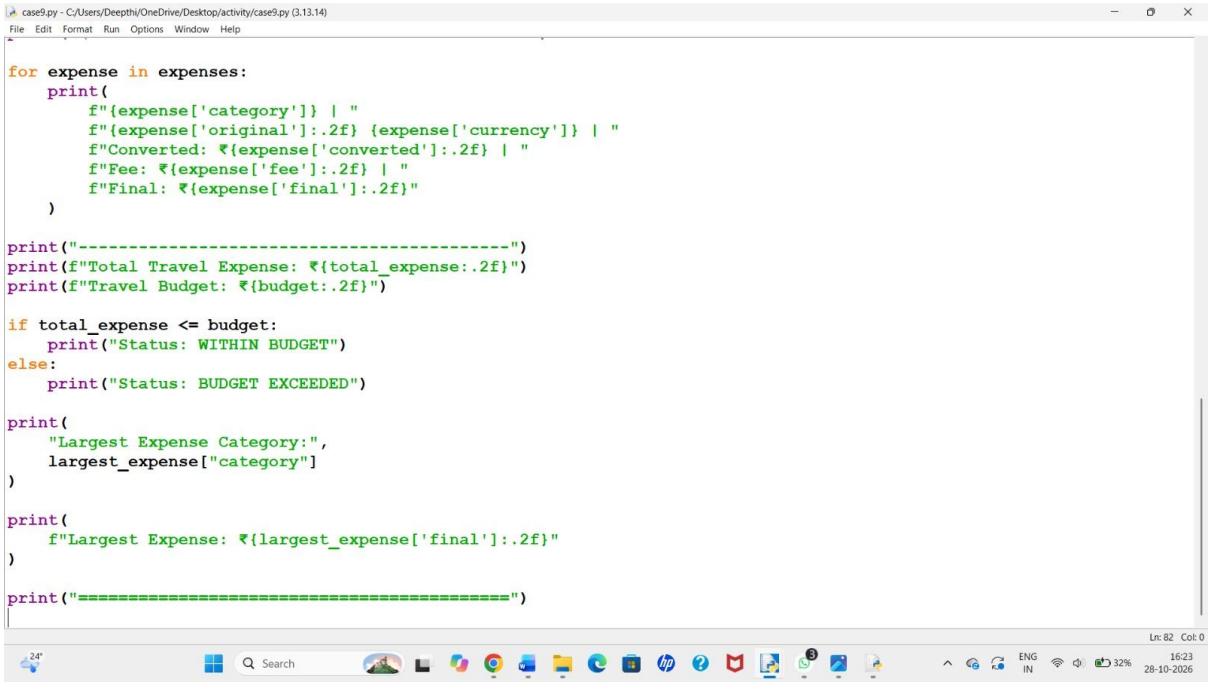
for i in range(n):
    print(f"\nExpense {i + 1}")

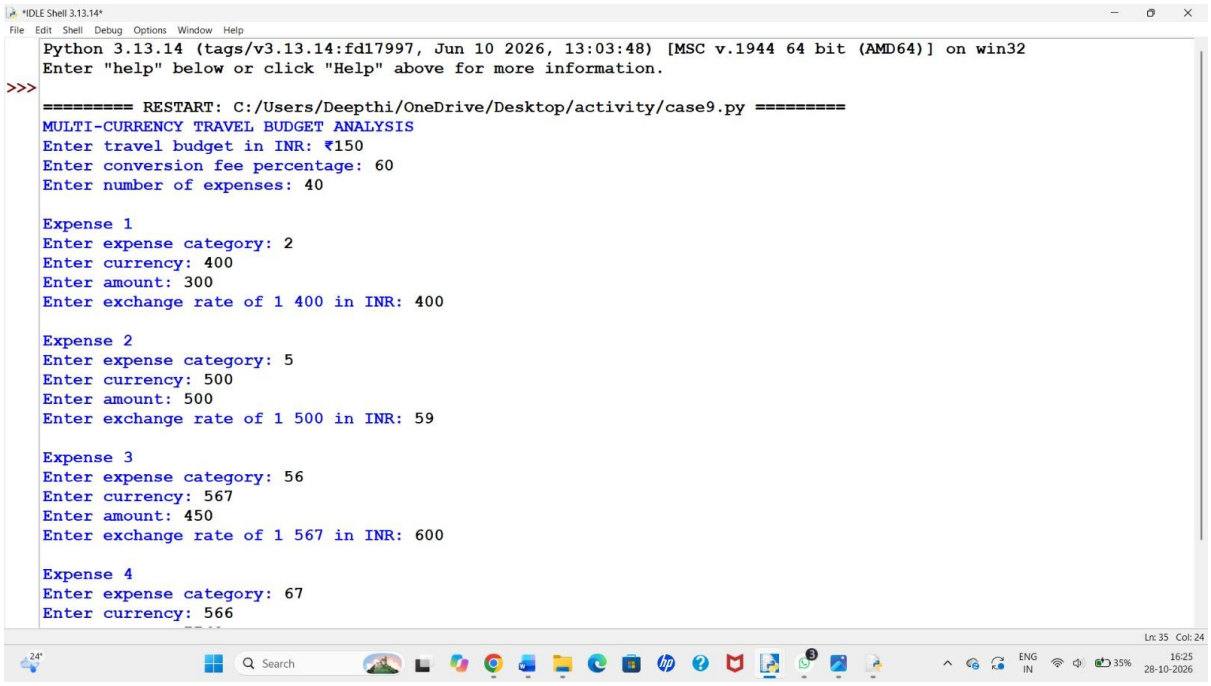
    category = input("Enter expense category: ")
    currency = input("Enter currency: ")
    amount = float(input("Enter amount: "))
    exchange_rate = float(
        input(f"Enter exchange rate of 1 {currency} in INR: ")
    )

    # Currency conversion
    converted_amount = amount * exchange_rate

    # Conversion fee
    conversion_fee = (
        converted_amount * conversion_fee_percentage / 100
    )
```


Edit
Screenshot (98).png
Open with
case9.py - C:/Users/Deepthi/OneDrive/Desktop/activity/case9.py (3.13.14)
File Edit Format Run Options Window Help
final_amount = converted_amount + conversion_fee
expenses.append({
"category": category,
"currency": currency,
"original": amount,
"converted": converted_amount,
"fee": conversion_fee,
"final": final_amount
})
total_expense = sum(
expense["final"] for expense in expenses
)
Find largest expense category
largest_expense = max(
expenses,
key=lambda x: x["final"]
)
print("\n========== TRAVEL EXPENSE REPORT ==========")
for expense in expenses:
print(
f"{expense['category']} | "
f"{expense['original']:.2f} {expense['currency']} | "
f"Converted: ₹{expense['converted']:.2f} | "
f"Fee: ₹{expense['fee']:.2f} | "
f"Final: ₹{expense['final']:.2f}"
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28-10-2026


case9.py - C:/Users/Deepthi/OneDrive/Desktop/activity/case9.py (3.13.14)
File Edit Format Run Options Window Help
for expense in expenses:
print(
f"{expense['category']} | "
f"{expense['original']:.2f} {expense['currency']} | "
f"Converted: ₹{expense['converted']:.2f} | "
f"Fee: ₹{expense['fee']:.2f} | "
f"Final: ₹{expense['final']:.2f}"
)
print("---")
print(f"Total Travel Expense: ₹{total_expense:.2f}")
print(f"Travel Budget: ₹{budget:.2f}")
if total_expense <= budget:
print("Status: WITHIN BUDGET")
else:
print("Status: BUDGET EXCEEDED")
print(
"Largest Expense Category:",
largest_expense["category"]
)
print(
f"Largest Expense: ₹{largest_expense['final']:.2f}"
)
print("===")
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28-10-2026



INPUT number of expense categories

expenses = empty list

FOR each category

INPUT category, foreign amount, exchange rate, fee percentage

converted = foreign amount * exchange rate

total cost = converted * (1 + fee percentage / 100)

STORE category and total cost

ENDFOR

INPUT budget

total spending = sum of all costs

largest expense = category with highest cost

DISPLAY costs, total, remaining budget, and largest category

Python Code

```
expenses = []

count = int(input("Number of expense categories: "))
```

```
for i in range(count):
    print(f"\nExpense {i + 1}")
    category = input("Category: ")
    foreign_amount = float(input("Amount in foreign currency: "))
    exchange_rate = float(input("Home currency per foreign unit: "))
    fee_percent = float(input("Conversion fee (%): "))

    if min(foreign_amount, exchange_rate, fee_percent) < 0:
        print("Invalid: amounts, rates, and fees cannot be negative.")
        raise SystemExit

    converted_amount = foreign_amount * exchange_rate
    total_cost = converted_amount * (1 + fee_percent / 100)
    expenses.append([category, total_cost])

budget = float(input("\nTravel budget in home currency: "))
if budget < 0:
    print("Budget cannot be negative.")
    raise SystemExit

total_spending = sum(item[1] for item in expenses)

for category, cost in expenses:
    print(f"{category}: {cost:.2f}")

print(f"\nTotal spending: {total_spending:.2f}")
print(f"Budget remaining: {budget - total_spending:.2f}")
```

```
print("Within budget." if total_spending <= budget else "Over budget.")


if expenses:
    largest = max(expenses, key=lambda item: item[1])
    print(f"Largest expense: {largest[0]} ({largest[1]:.2f})")
else:
    print("No expense categories were entered.")
```

Test Case and Expected Output

Hotel: 100 foreign units at 80 home currency per unit, with 2% fee. Food: 50 foreign units at the same rate, with 1% fee. Budget: 13,000.

Hotel: 8160.00

Food: 4040.00

Total spending: 12200.00

Budget remaining: 800.00

Within budget.

Largest expense: Hotel (8160.00)

Conclusion

The program converts each expense, includes its fee, compares total spending with the budget, and reports the largest cost category.

Case 10: Campus Event Financial Planning

Problem Statement

Calculate event costs for venue, equipment, catering, transport, and certificates. Determine participant contributions, sponsorship shortfall, and the minimum registrations needed to break even.

Objective

Calculate total event costs and funding, identify any shortfall, and estimate the break even registration count.

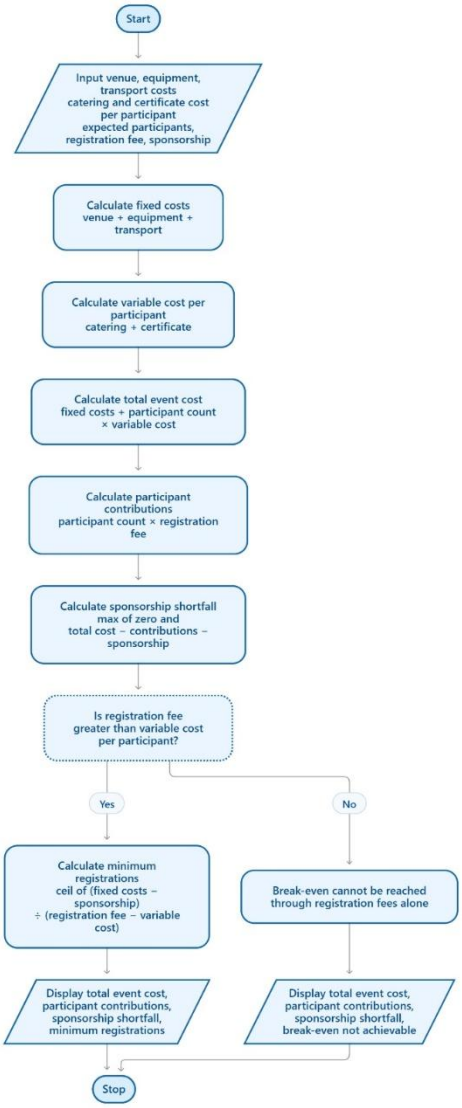
Assumptions

- Venue, equipment, and transport are fixed costs.
- Catering and certificate costs are per participant.
- Each participant pays the entered registration fee.
- Sponsorship is a fixed contribution.
- Break even registrations are calculated from total fixed costs after sponsorship and the contribution remaining after per participant costs.

Algorithm

1. Read fixed and per participant costs.
2. Read participant count, registration fee, and sponsorship.
3. Calculate the event's total cost and participant contributions.
4. Calculate total funding and any shortfall.
5. Calculate the minimum registrations needed to cover costs.
6. Display the results.

Flowchart



Pseudocode

case10.py - C:/Users/Deepthi/OneDrive/Desktop/activity/case10.py (3.13.14)

File Edit Format Run Options Window Help

```
venue = float(input("Enter venue cost: "))
equipment = float(input("Enter equipment cost: "))
catering = float(input("Enter catering cost: "))
transport = float(input("Enter transport cost: "))
certificates = float(input("Enter certificate cost: "))

participants = int(input("Enter number of participants: "))
fee = float(input("Enter registration fee per participant: "))
sponsorship = float(input("Enter sponsorship amount: "))

# Calculate total cost
total_cost = venue + equipment + catering + transport + certificates

# Calculate participant contribution
contribution = participants * fee

# Calculate shortfall
shortfall = total_cost - contribution - sponsorship

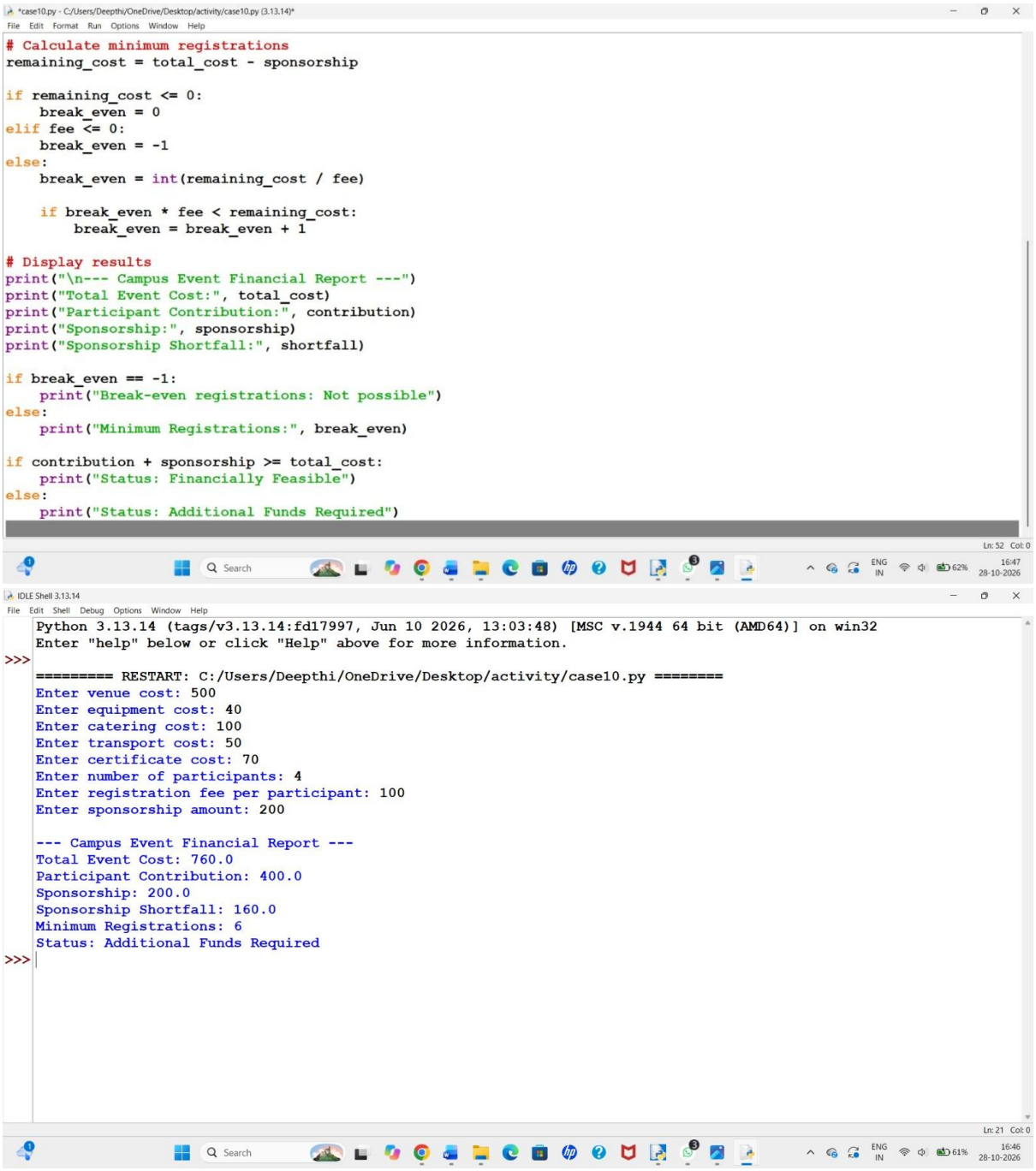
if shortfall < 0:
    shortfall = 0

# Calculate minimum registrations
remaining_cost = total_cost - sponsorship

if remaining_cost <= 0:
    break_even = 0
elif fee <= 0:
    break_even = -1
else:
```

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```
# Calculate minimum registrations
remaining_cost = total_cost - sponsorship

if remaining_cost <= 0:
    break_even = 0
elif fee <= 0:
    break_even = -1
else:
    break_even = int(remaining_cost / fee)

    if break_even * fee < remaining_cost:
        break_even = break_even + 1

# Display results
print("\n--- Campus Event Financial Report ---")
print("Total Event Cost:", total_cost)
print("Participant Contribution:", contribution)
print("Sponsorship:", sponsorship)
print("Sponsorship Shortfall:", shortfall)

if break_even == -1:
    print("Break-even registrations: Not possible")
else:
    print("Minimum Registrations:", break_even)

if contribution + sponsorship >= total_cost:
    print("Status: Financially Feasible")
else:
    print("Status: Additional Funds Required")
```

```
Python 3.13.14 (tags/v3.13.14:fd17997, Jun 10 2026, 13:03:48) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
========= RESTART: C:/Users/Deepthi/OneDrive/Desktop/activity/case10.py ========
Enter venue cost: 500
Enter equipment cost: 40
Enter catering cost: 100
Enter transport cost: 50
Enter certificate cost: 70
Enter number of participants: 4
Enter registration fee per participant: 100
Enter sponsorship amount: 200

--- Campus Event Financial Report ---
Total Event Cost: 760.0
Participant Contribution: 400.0
Sponsorship: 200.0
Sponsorship Shortfall: 160.0
Minimum Registrations: 6
Status: Additional Funds Required
>>>
```

INPUT venue, equipment, transport

INPUT catering and certificate costs per participant

INPUT expected participants, registration fee, sponsorship

fixed cost = venue + equipment + transport

variable cost per participant = catering + certificate

total cost = fixed cost + participants * variable cost per participant

```
contributions = participants * registration fee
total funding = contributions + sponsorship
shortfall = maximum(0, total cost - total funding)

IF registration fee <= variable cost per participant
   break even registrations = not achievable from registration alone
ELSE
   break even registrations =
      ceiling((fixed cost - sponsorship) /
            (registration fee - variable cost per participant))
ENDIF

DISPLAY costs, funding, shortfall, and break even count
```

Python Code

```
import math

venue = float(input("Venue cost: "))
equipment = float(input("Equipment cost: "))
transport = float(input("Transport cost: "))
catering_per_person = float(input("Catering cost per participant: "))
certificate_per_person = float(input("Certificate cost per participant: "))
participants = int(input("Expected number of participants: "))
registration_fee = float(input("Registration fee per participant: "))
sponsorship = float(input("Sponsorship received: "))

values = [
   venue, equipment, transport, catering_per_person,
   certificate_per_person, participants, registration_fee, sponsorship
```

```
]

if min(values) < 0:
    print("Invalid: costs, participant count, and funding cannot be negative.")
else:
    fixed_cost = venue + equipment + transport
    variable_cost = catering_per_person + certificate_per_person
    total_cost = fixed_cost + participants * variable_cost
    participant_contributions = participants * registration_fee
    total_funding = participant_contributions + sponsorship
    shortfall = max(0, total_cost - total_funding)

    amount_needed_from_registrations = fixed_cost - sponsorship
    contribution_per_registration = registration_fee - variable_cost

    if amount_needed_from_registrations <= 0:
        break_even = 0
    elif contribution_per_registration <= 0:
        break_even = None
    else:
        break_even = math.ceil(
            amount_needed_from_registrations / contribution_per_registration
        )

    print(f"\nFixed costs: {fixed_cost:.2f}")
    print(f"Total event cost: {total_cost:.2f}")
    print(f"Participant contributions: {participant_contributions:.2f}")
    print(f"Sponsorship: {sponsorship:.2f}")
```

```
print(f"Overall funding shortfall: {shortfall:.2f}")


if break_even is None:
    print("Break even is not achievable from registration fees alone.")
else:
    print(f"Minimum registrations to break even: {break_even}")
```

Test Case and Expected Output

Fixed costs: venue 5,000; equipment 1,000; transport 1,000. Catering and certificate costs are 100 per participant. Registration fee: 200. Sponsorship: 1,000. Expected participants: 50.

Fixed costs: 7000.00

Total event cost: 12000.00

Participant contributions: 10000.00

Sponsorship: 1000.00

Overall funding shortfall: 1000.00

Minimum registrations to break even: 60

Conclusion

The program calculates fixed and variable event costs, totals the available funding, finds any shortfall, and estimates the registrations needed to break even.